# Applied AI Mastery Program

# Module 1 – From Linear Models to Deep Learning

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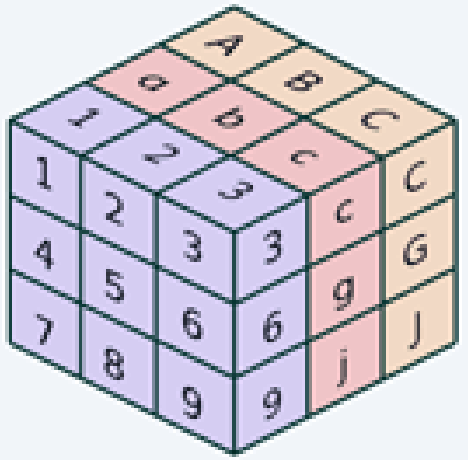
# Recap – Scalar, Vector, Matrix, Tensor

| Scalar | Row Vector (shape 1×3) | Column Vector (shape 3×1) | Matrix | Tensor |
|---|---|---|---|---|
| (11) | 5 3 7 | 5<br>1.5<br>2 | 4 19 8<br>16 3 5 |  |
| (,) | (1,3) | (3,1) | (2,3) | (3,3,3) |

# Physical Properties as a Scalar or Vector

## Scalar Quantities

length, area, volume
speed
mass, density
pressure
temperature
energy, entropy
work, power


## Vector Quantities

displacement, direction
velocity
acceleration
momentum
force
lift , drag , thrust
weight

# Linear Equations

Matrix multiplation

$$y = \mathbf{W}\mathbf{X}^{T} + w_0$$

(,) (1,m) (m,1) (,)

(1,1)=(,)

$$\mathbf{W} = [[w_1, \ldots, w_m]] \qquad \mathbf{X} = [[x_1, \ldots, x_m]]$$

# Linear Equations

$$y = \overset{(1,m)}{[w_1, \dots, w_m]} \overset{(m,1)}{\begin{bmatrix} x_1 \\ \dots \\ x_m \end{bmatrix}} + w_0$$

$$= \sum_{i=1}^{m} w_i x_i + w_0$$

# Linear Equations

$$y = w_1 x_{qa} + w_2 x_{at} + w_3 x_{hw} + w_4 x_{te} + w_0$$

The input features (components of the feature vector $x$) are:

- $x_{qa}$ —number of questions answered in class
- $x_{at}$ —attendance rate
- $x_{hw}$ —homework score
- $x_{te}$ —midterm exam score

for one sample

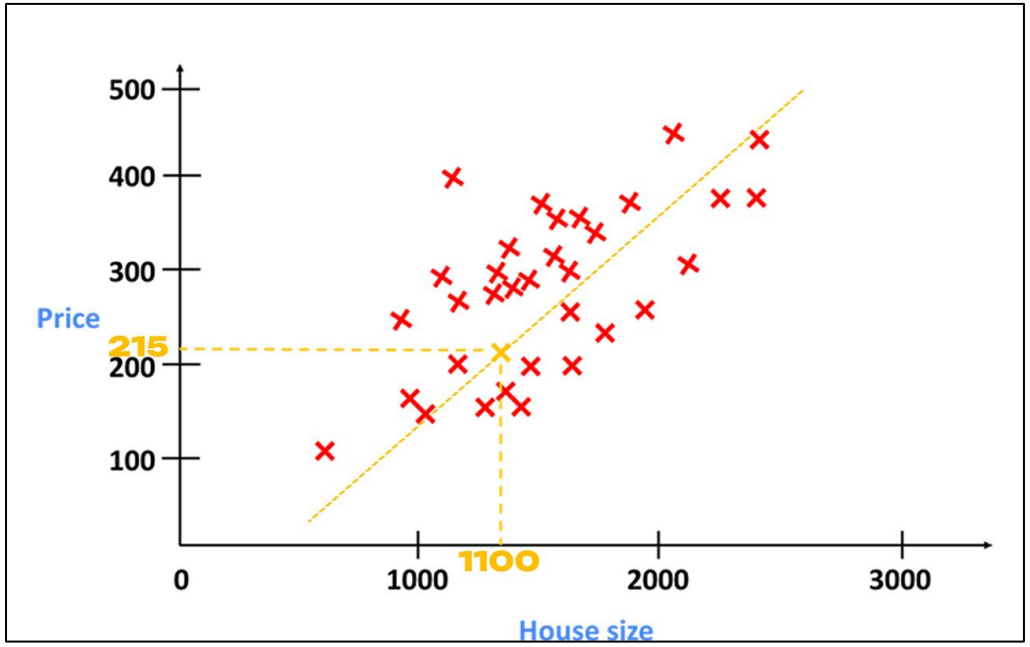
$$\mathbf{W} = [[w_1, w_2, w_3, w_4]] \quad \mathbf{X} = [[x_{qa}, x_{at}, x_{hw}, x_{te}]]$$

# Linear Equations

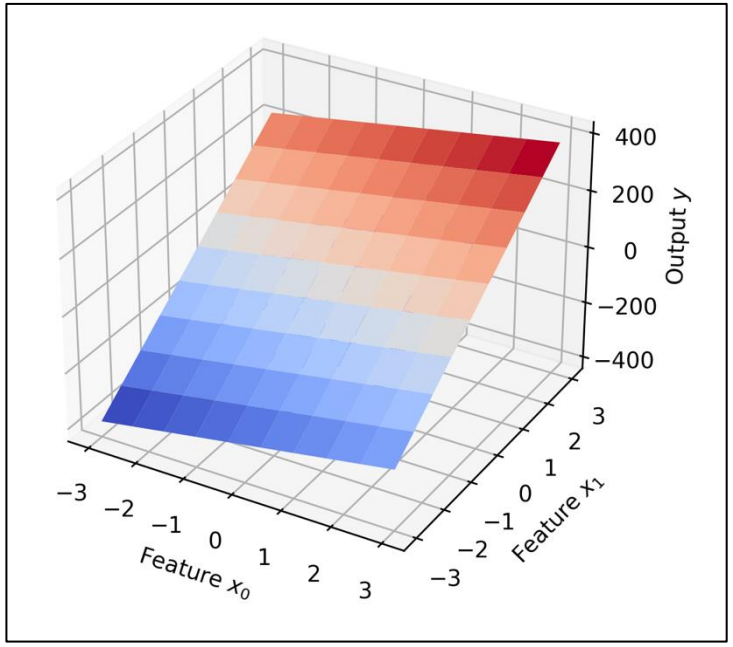
$$y = \boldsymbol{w_1 x_1} + w_0$$

For example, $y = w_1 x_{qa} + w_o$

# 1D vs. 2D vs. $n$ −Dimension




A Line




A plane

How about $n-dimension$

A hyper-plane

# Linear Equations

$$y = w_1 x_{qa} + w_2 x_{at} + w_3 x_{hw} + w_4 x_{te} + w_0$$

***Matrix form***

$$\underset{(,)}{y} = \underset{(1,m)\ (m,1)}{\mathbf{W}\mathbf{X}^T} + \underset{(,)}{w_0}$$

(1,1)=(,)

***Many n samples, one output (scalar)***

$$\underset{(1,n)}{y} = \underset{(1,m)\ (m,n)}{\mathbf{W}\mathbf{X}^T} + \underset{(,)}{w_0}$$

(1,n)

*Broadcasting $w_0$ to be (1,n) by cloning*

# Broadcasting

**5**

(,)

*Broadcasting a scalar to a column vector (1,5)*


(1,5)

# Linear Equations

$$y = \mathbf{W}\mathbf{X}^{T} + w_0$$

(,) (1,m) (m,1) (,)

(1,1)=(,)

***Many $n$ samples***

$$y = \mathbf{W}\mathbf{X}^{T} + w_0$$

(1,n) (1,m) (m,n) (,)

(1,n)

*Broadcasting $w_0$ to be (1,n) by cloning*

***Many $n$ samples, many $k$ outputs***

$$\mathbf{Y} = \mathbf{W}\mathbf{X}^{T} + w_0$$

(k,n) (k,m) (m,n) (k,)

(k,n)

# Example

$\mathbf{X}$ (4,3)

| Sample | Feature 1 | Feature 2 | Feature 3 |
|---|---|---|---|
| $x^{(1)}$ | 2 | 4 | 6 |
| $x^{(2)}$ | 5 | 1 | 3 |
| $x^{(3)}$ | 1 | 2 | 8 |
| $x^{(4)}$ | 3 | 7 | 4 |

$\mathbf{W}$ (2,3)

$$\begin{bmatrix} 0.5 & 0.2 & 0.1 \\ 1.0 & 0.3 & 0.4 \end{bmatrix}$$

$w_0$ (2,)

$$\begin{bmatrix} 10 \\ 20 \end{bmatrix}$$

# Example

**X** (4,4)

| Sample | Bias Feature | Feature 1 | Feature 2 | Feature 3 |
|---|---|---|---|---|
| $x^{(1)}$ | 1 | 2 | 4 | 6 |
| $x^{(2)}$ | 1 | 5 | 1 | 3 |
| $x^{(3)}$ | 1 | 1 | 2 | 8 |
| $x^{(4)}$ | 1 | 3 | 7 | 4 |

**W** (2,4)

***Augmented columns***

***Depending our libraries, this part is usually done for you behind the scene.***

$$\tilde{W} = \begin{bmatrix} w_0 & W \end{bmatrix} = \begin{bmatrix} 10 & 0.5 & 0.2 & 0.1 \\ 20 & 1.0 & 0.3 & 0.4 \end{bmatrix}$$

# Linear Equations

$$y = \mathbf{W}\mathbf{X}^T$$

(,) (1,m) (m,1) → (1,1)=(,)

***Many $n$ samples***

$$y = \mathbf{W}\mathbf{X}^T$$

(1,n) (1,m) (m,n) → (1,n)

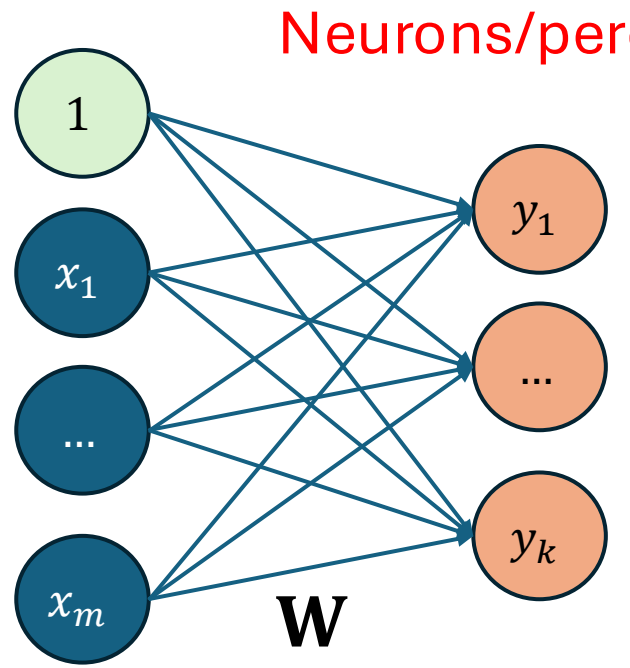
*Broadcasting $w_0$ to be (1,n) by cloning*

***Many $n$ samples, many $k$ outputs***

$$\mathbf{Y} = \mathbf{W}\mathbf{X}^T$$

(k,n) (k,m) (m,n) → (k,n)

# Computational Graph for One Sample

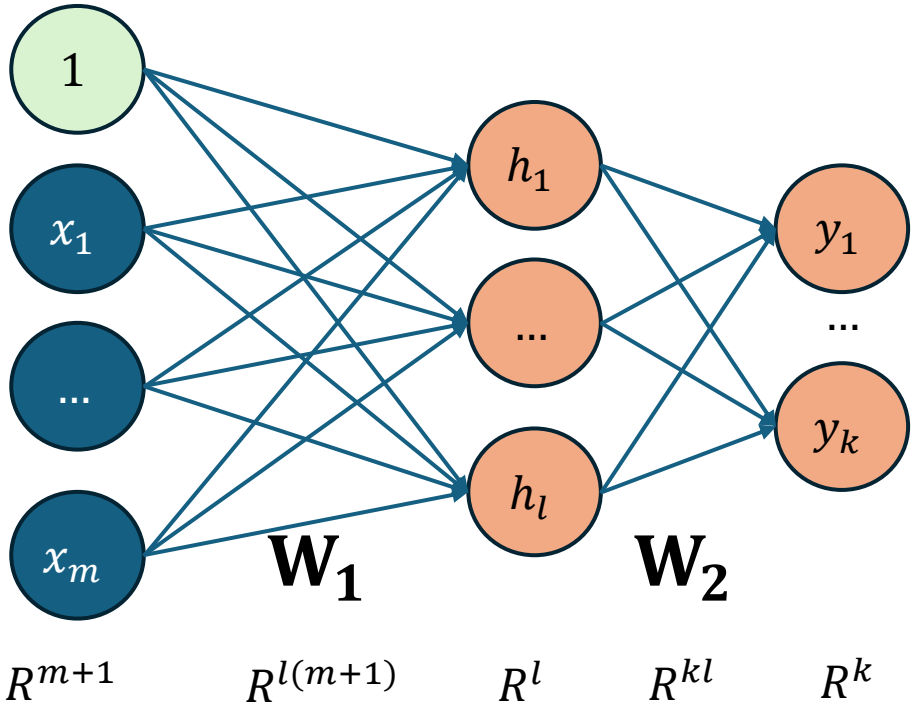



Linear Layer or Dense Layer or Feedforward Layer

$$\mathbf{Y} = \mathbf{W}\mathbf{X}^T$$

Linear Model

# Computational Graph for One Sample




Simple Neural Network or Multi-layered Perceptron Network

$$\mathbf{Y} = \mathbf{W_2}(\mathbf{W_1}\mathbf{X}^T)$$

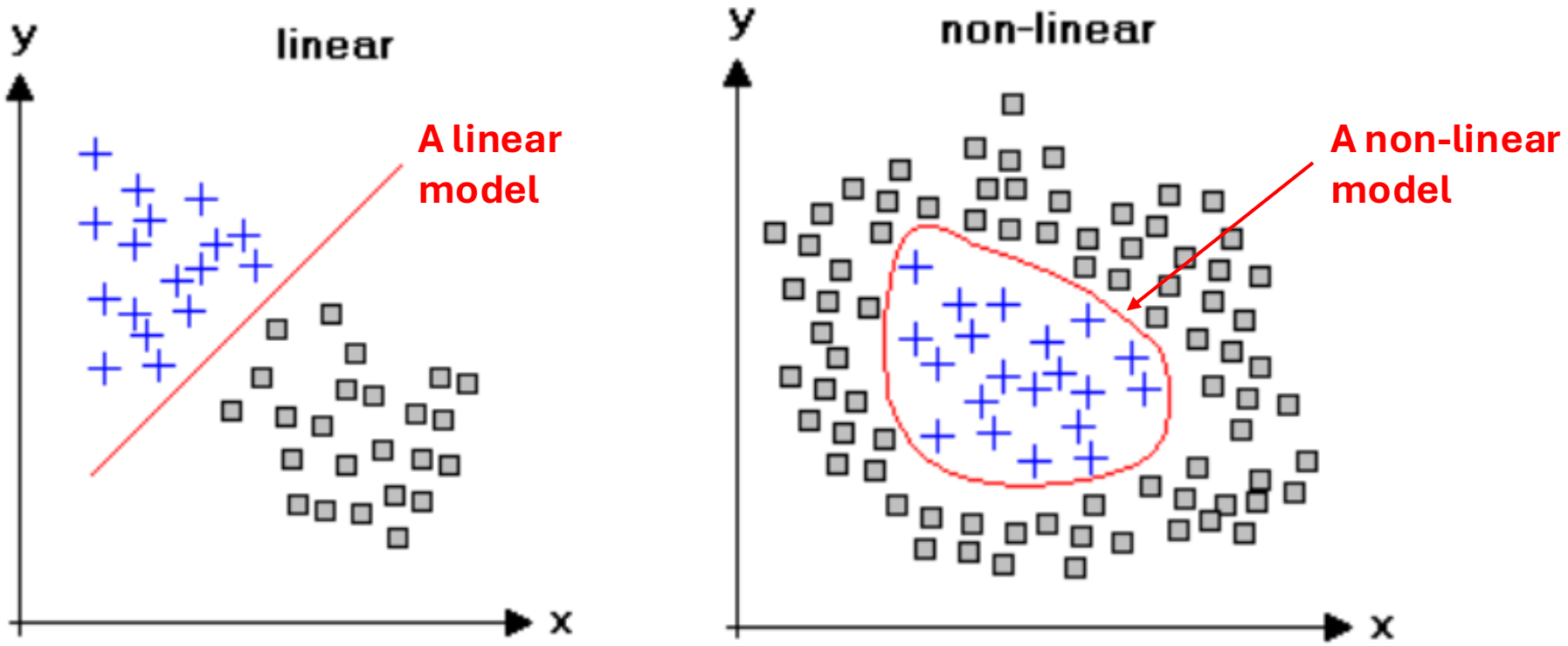
Linear Model

# Linear Definition

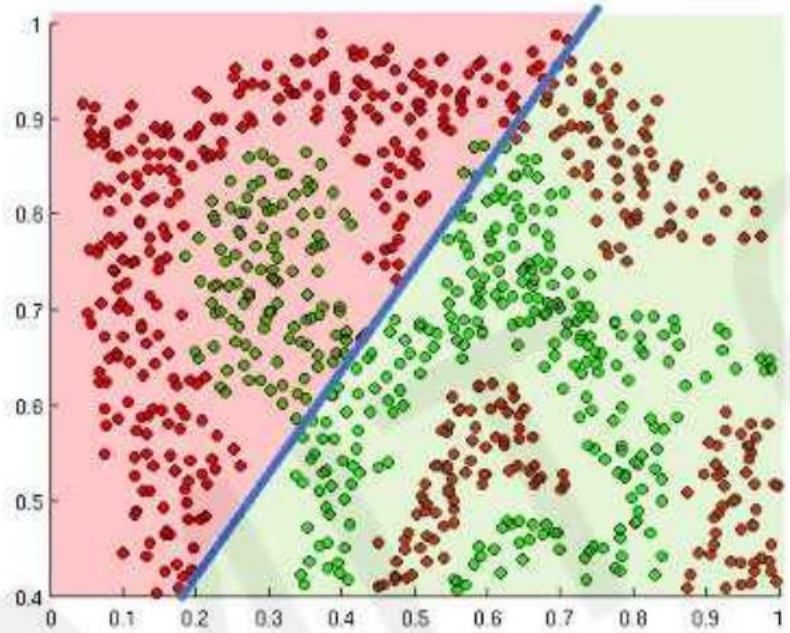
A linear function :

1. addition: $f(x+y) = f(x) + f(y)$
2. scaling: $f(ax) = af(x)$

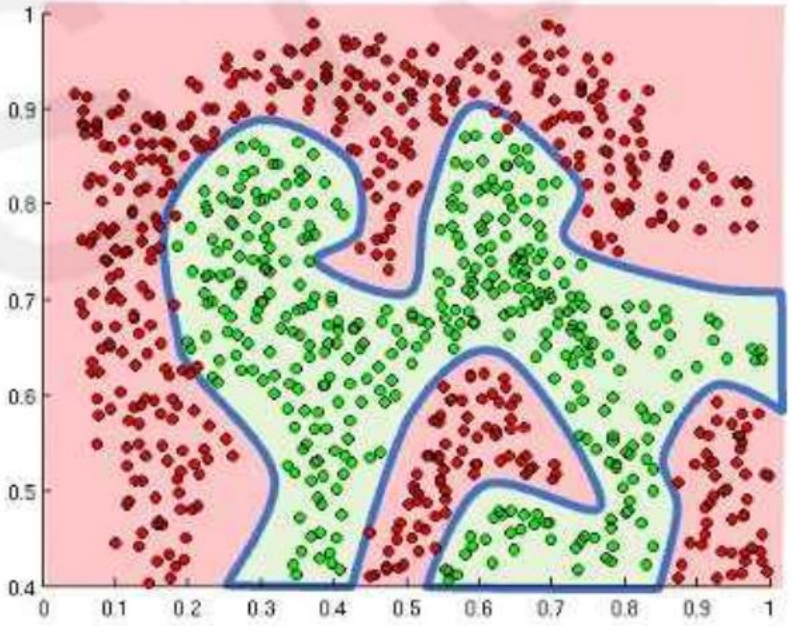
# Linearity vs. Non-Linearity



# How to Introduce Non-Linearities

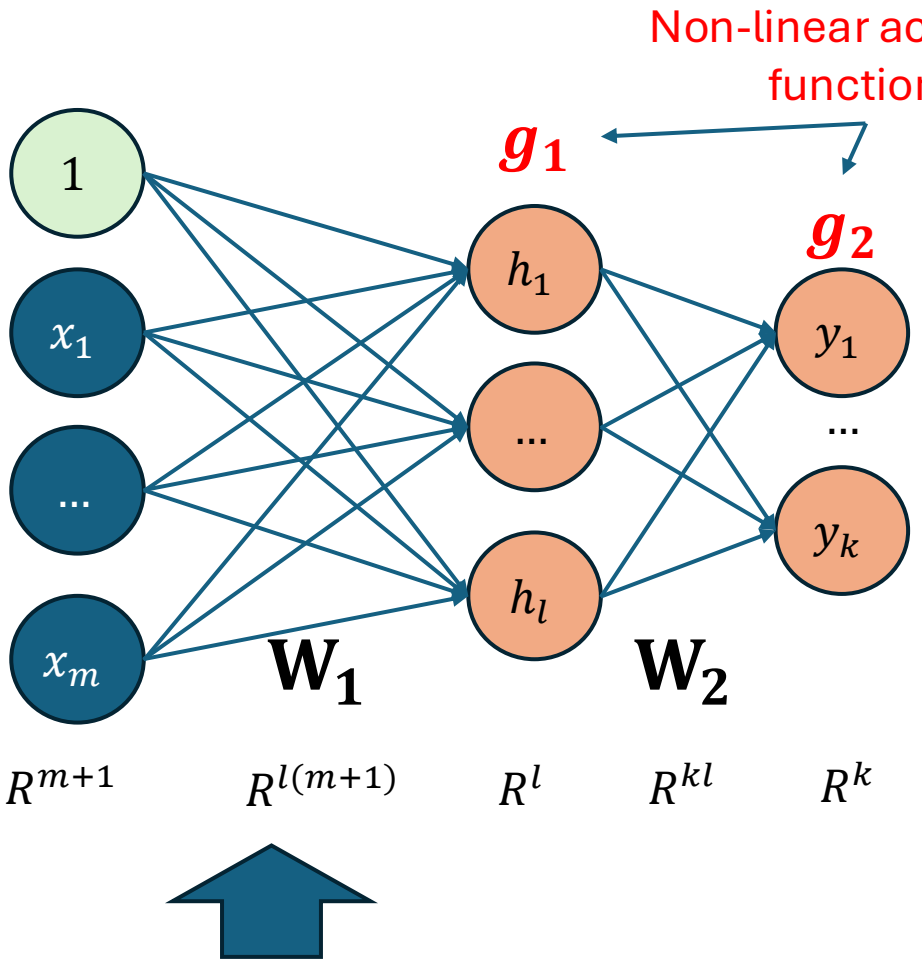


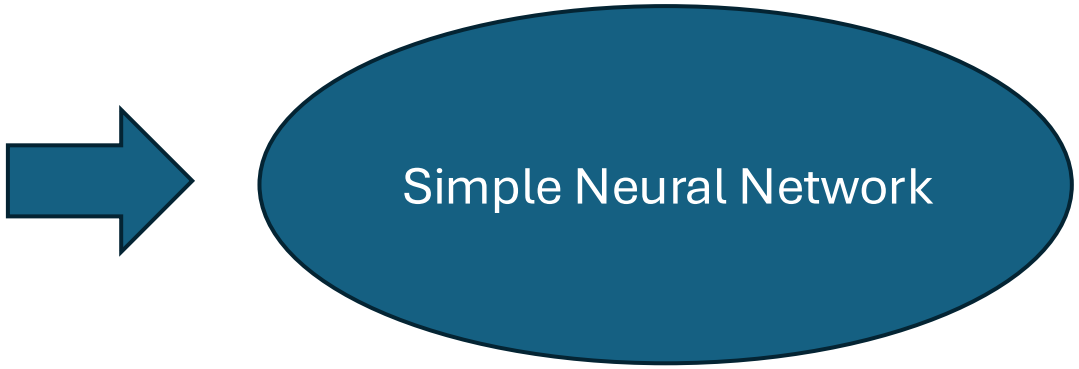

Linear activation functions produce linear decisions no matter the network size




Non-linearities allow us to approximate arbitrarily complex functions

# Computational Graph for One Sample



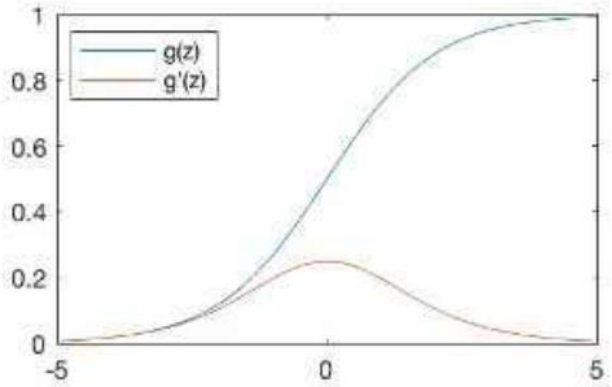




$$\mathbf{Y} = \boldsymbol{g_2}(\mathbf{W_2}\boldsymbol{g_1}(\mathbf{W_1}\mathbf{X}^{\boldsymbol{T}}))$$

Non-Linear Model

# Common Activation Functions

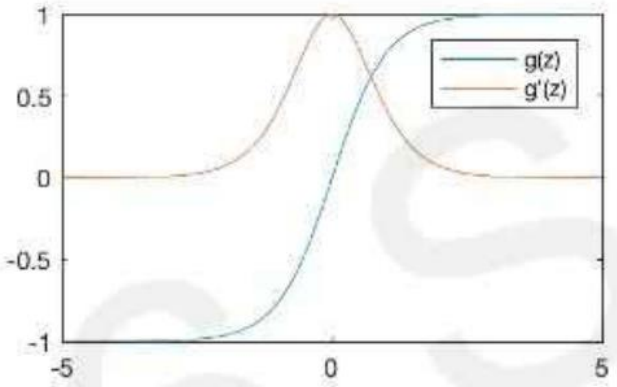
## Sigmoid Function




$$g(z) = \frac{1}{1 + e^{-z}}$$

$$g'(z) = g(z)(1 - g(z))$$

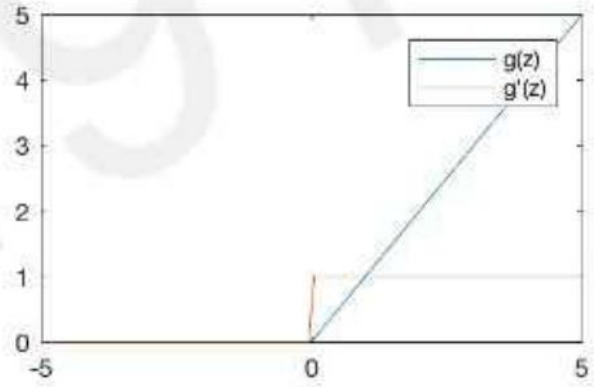
## Hyperbolic Tangent




$$g(z) = \frac{e^{z} - e^{-z}}{e^{z} + e^{-z}}$$

$$g'(z) = 1 - g(z)^2$$

## Rectified Linear Unit (ReLU)

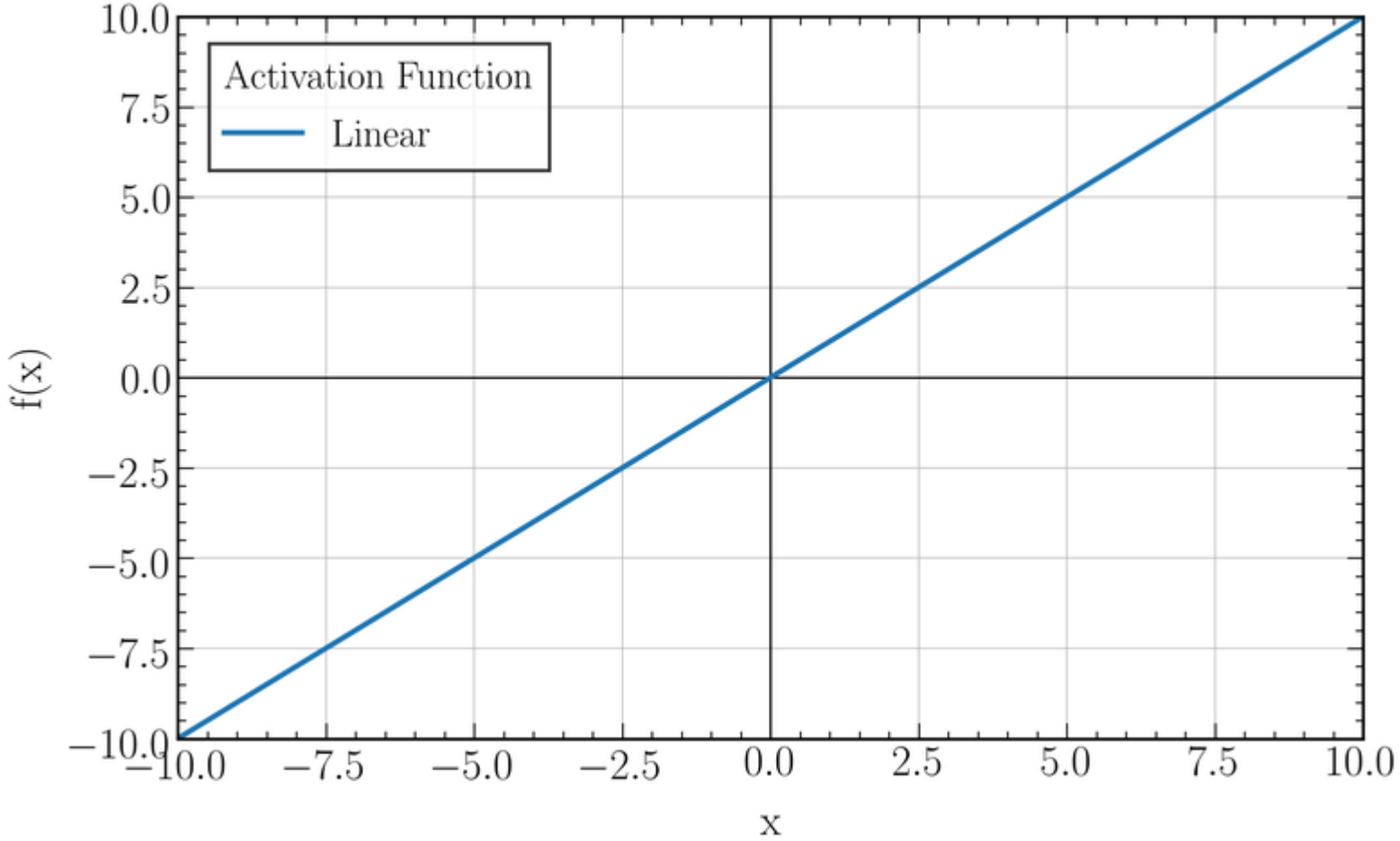



$$g(z) = \max(0, z)$$

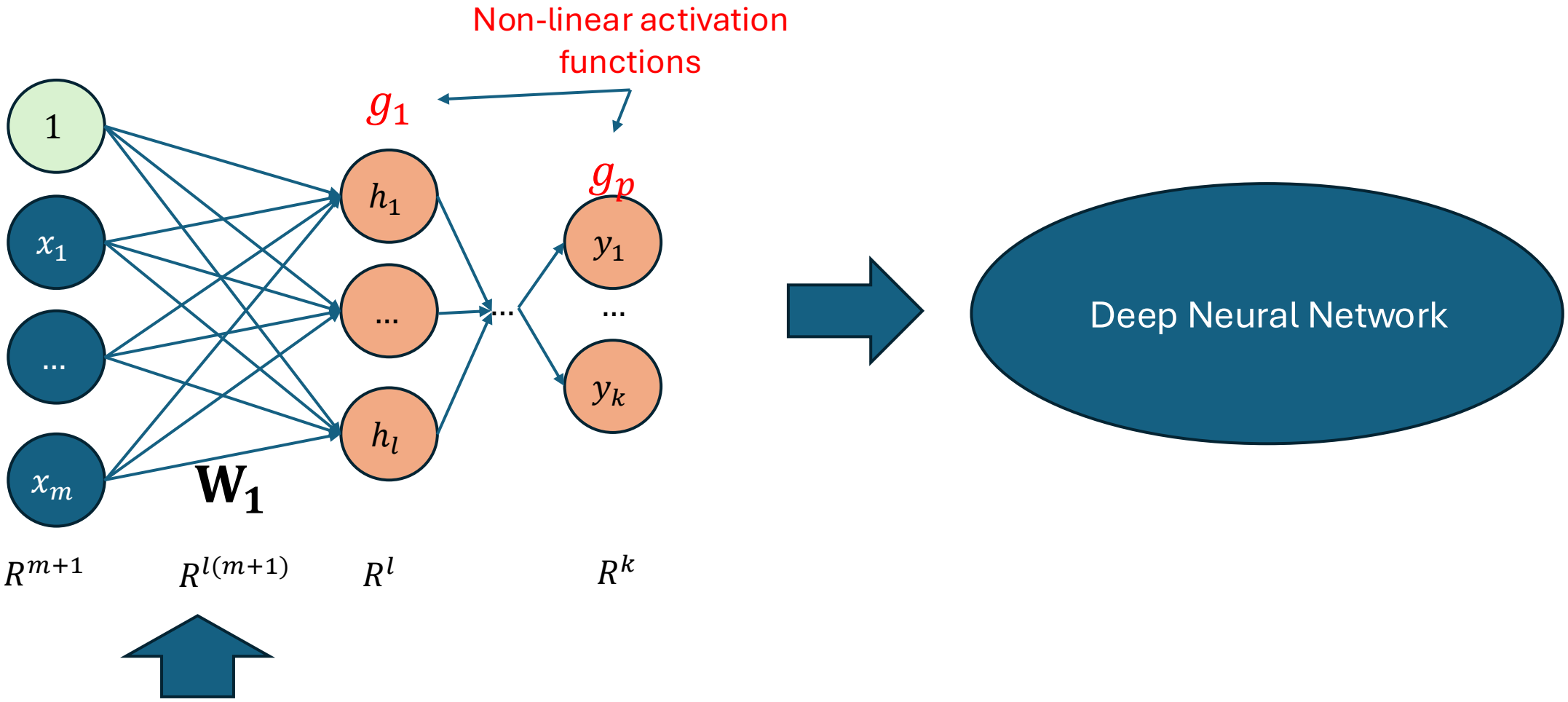
$$g'(z) = \begin{cases} 1, & z > 0 \\ 0, & \text{otherwise} \end{cases}$$

# Common Activation Functions

$$f(x) = x$$



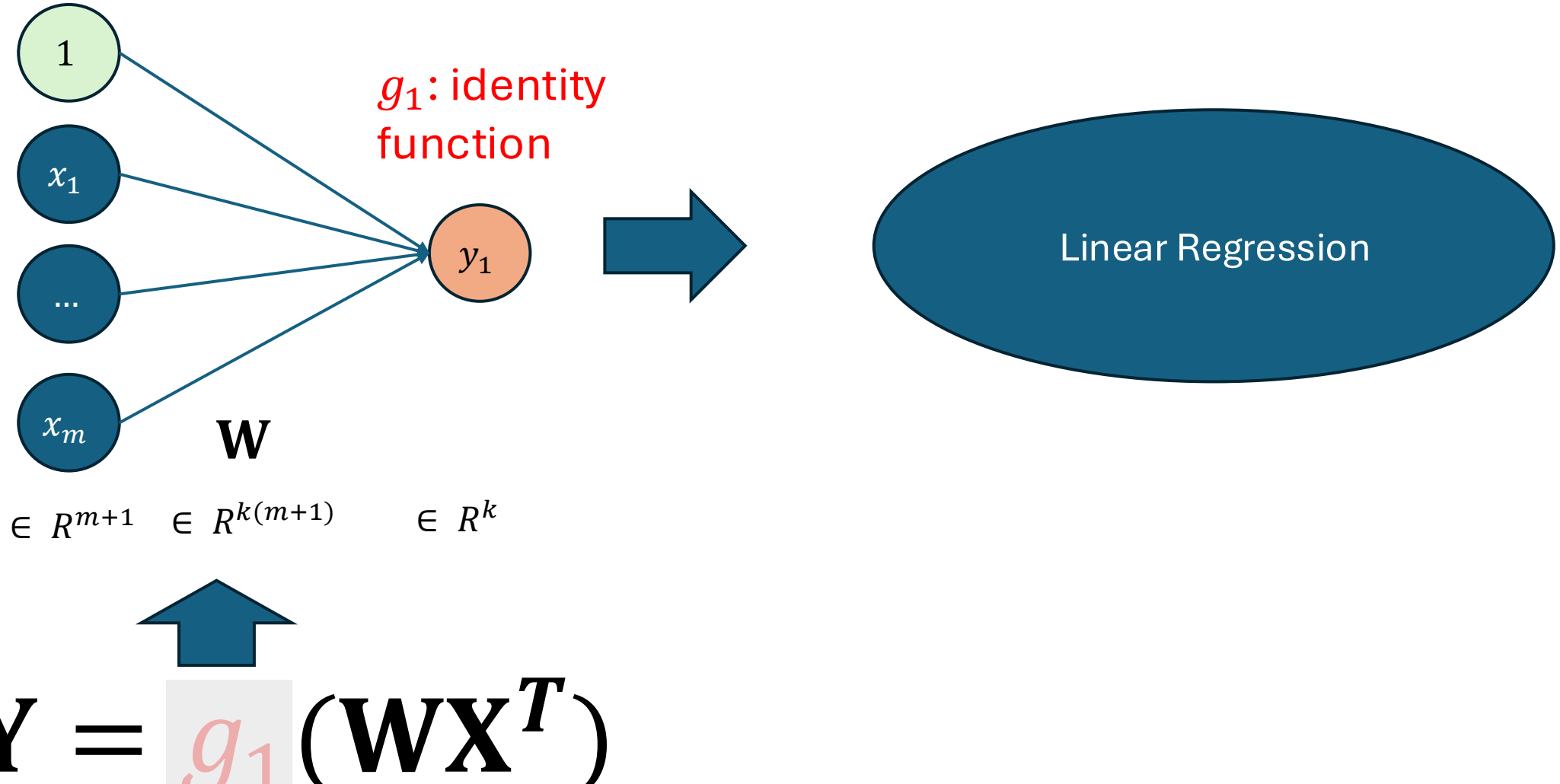
# Computational Graph for One Sample



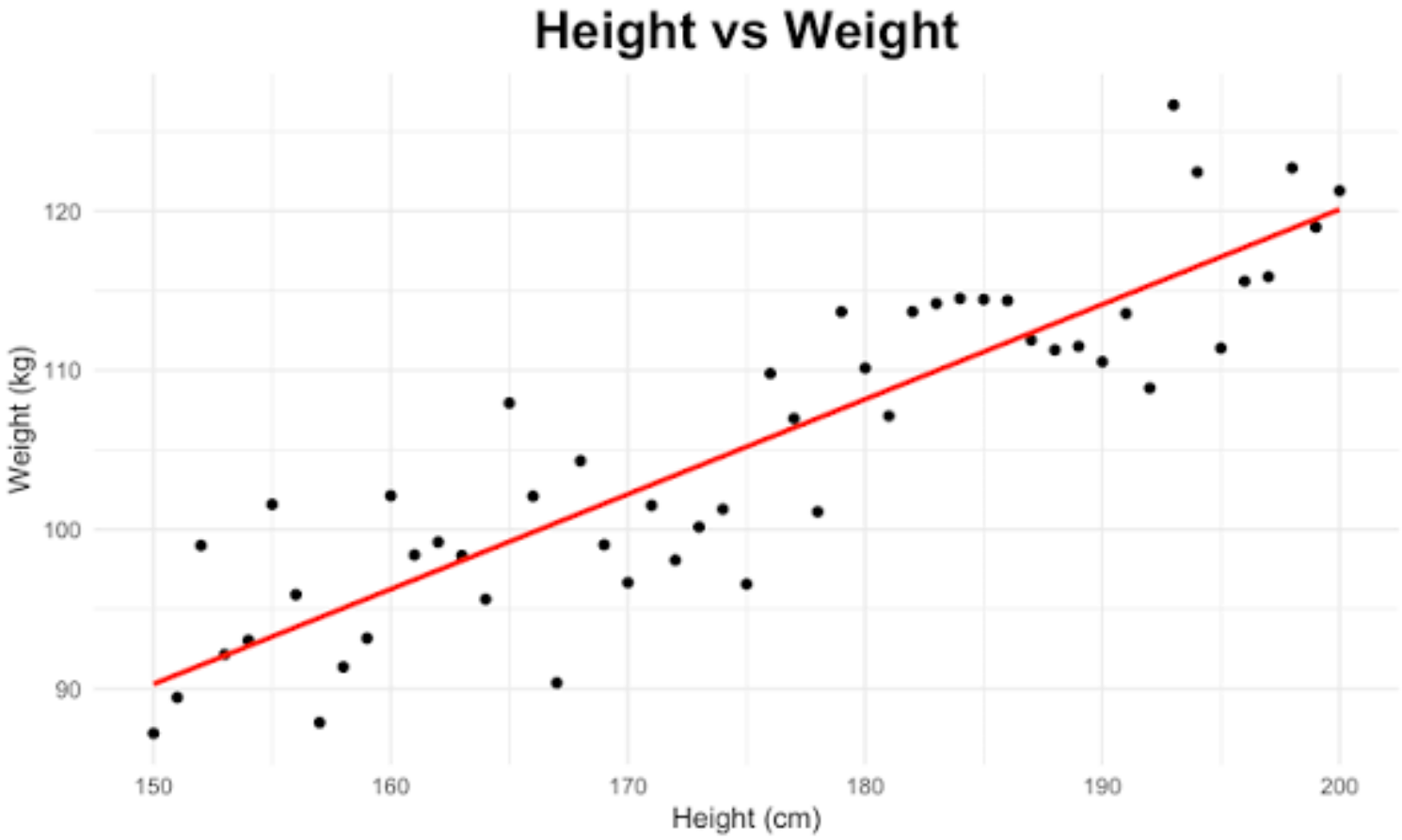

$$\mathbf{Y} = g_p(\dots g_1(\mathbf{W_1 X}^T))$$

Non-Linear Model

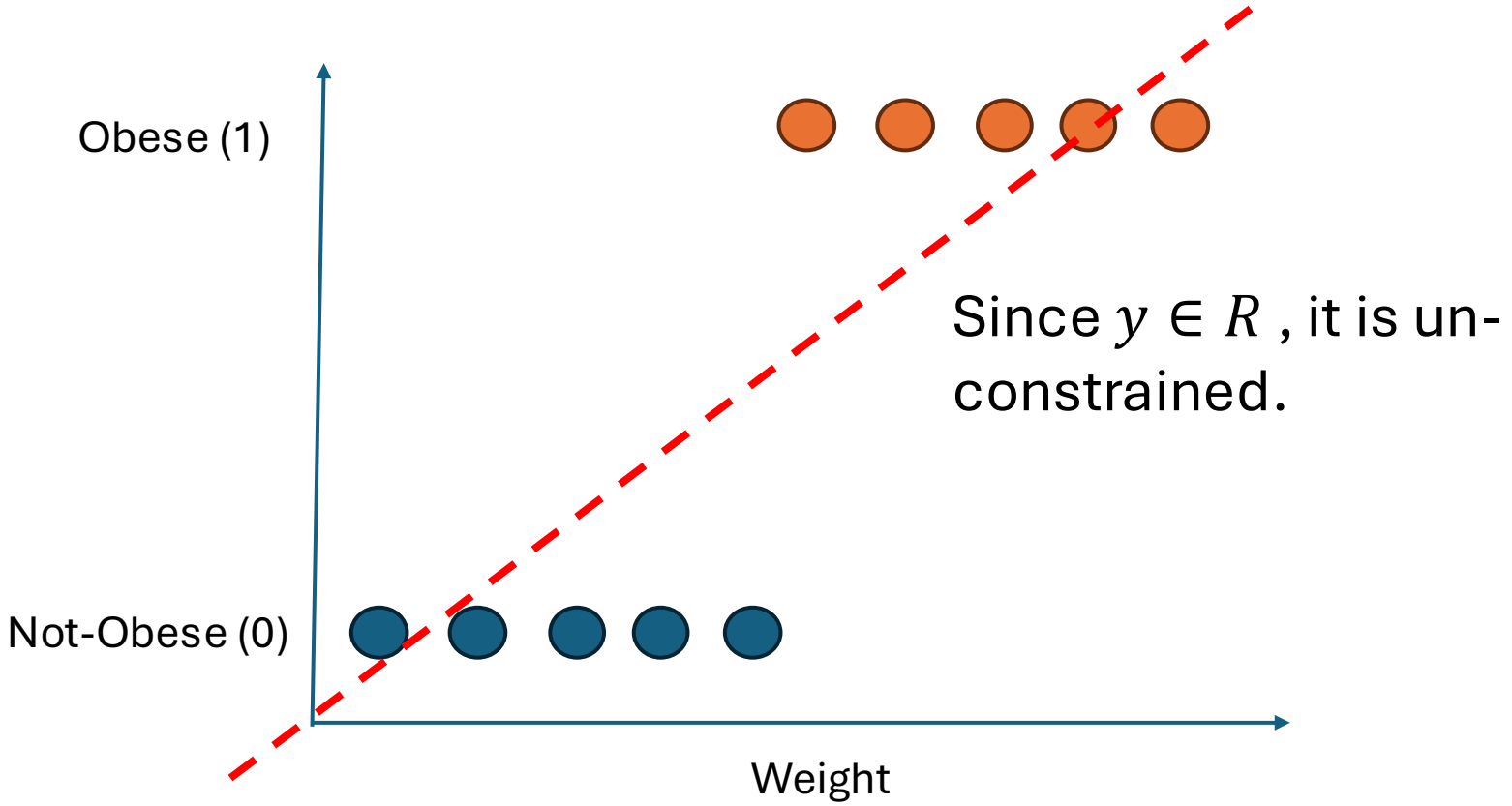
# Linear Regression as a Special Linear Layer



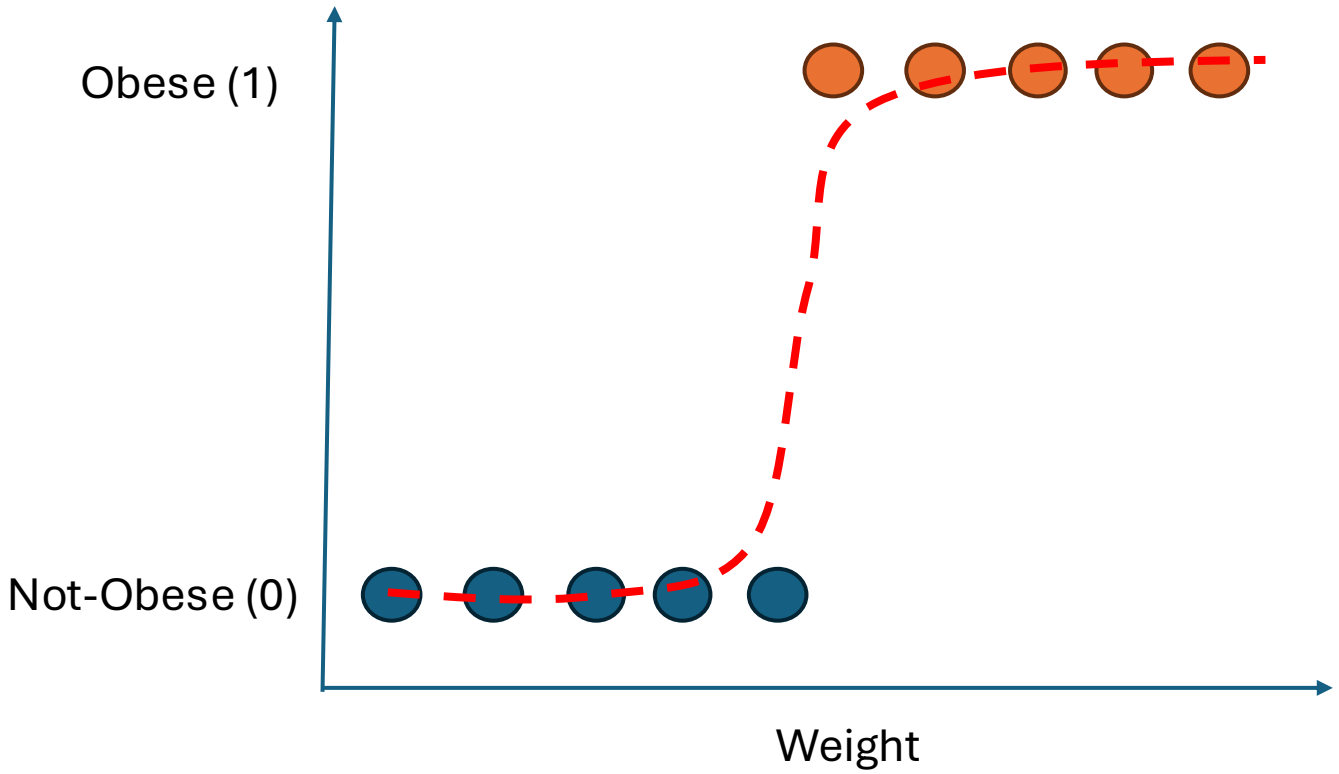
# Linear Regression as a Special Linear Layer



# Logistic Regression as a Special Linear Layer



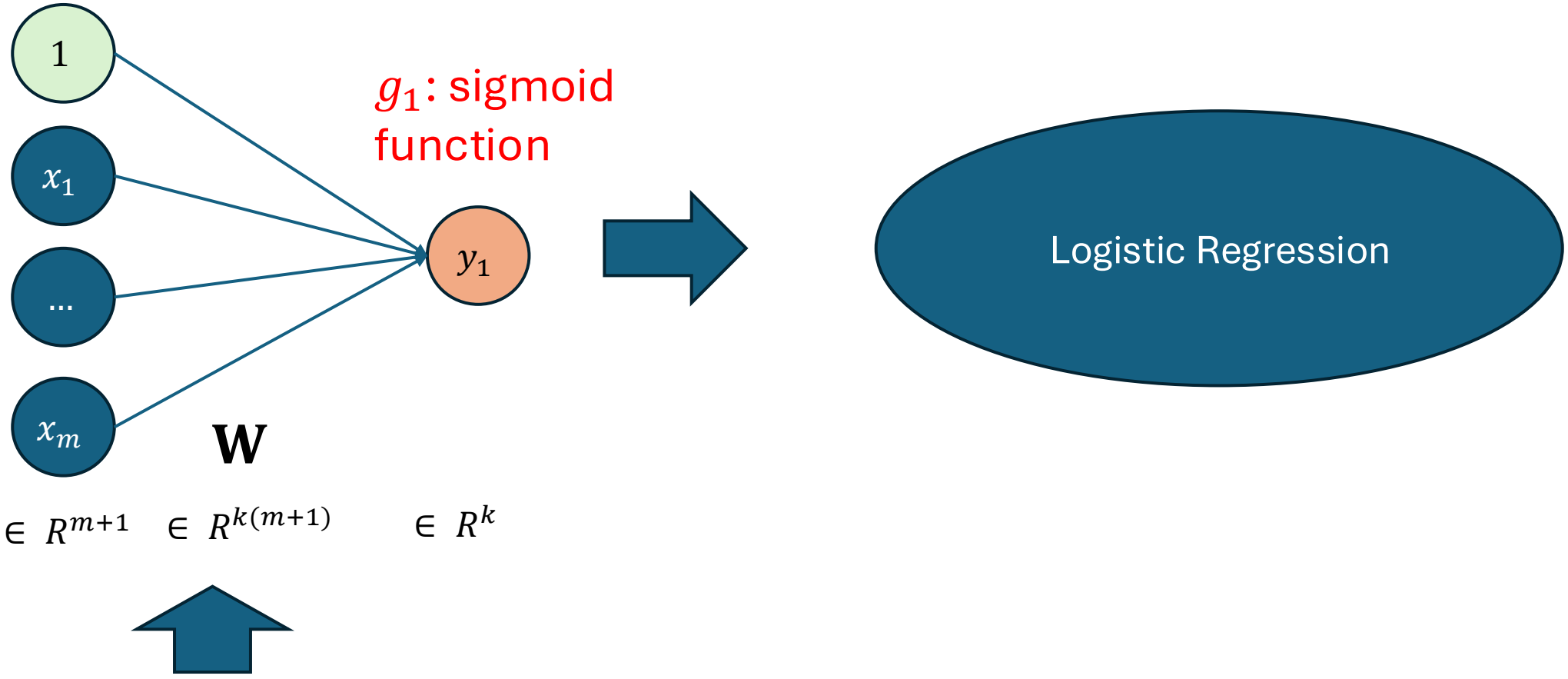
# Logistic Regression as a Special Linear Layer



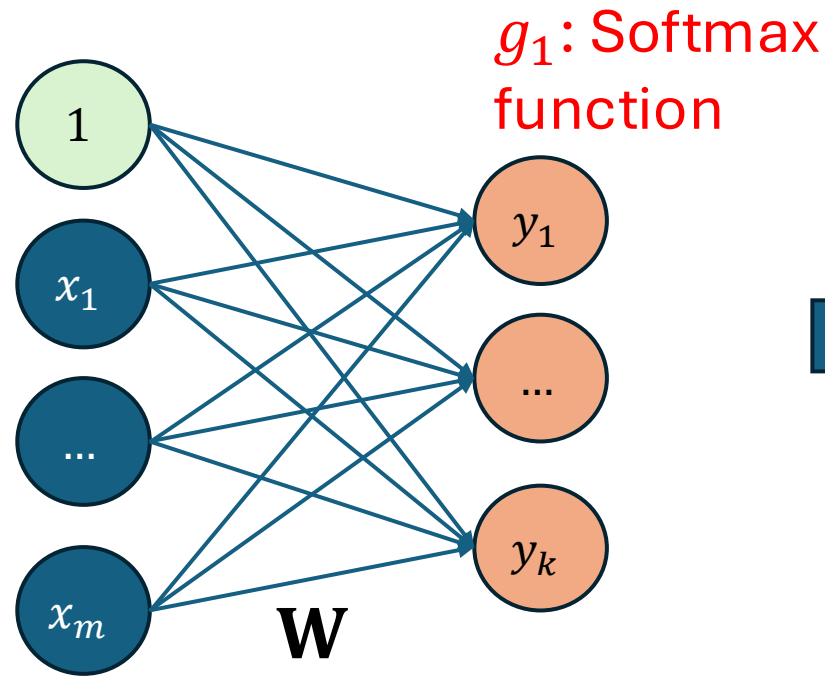

What if we predict $p(obese\ or\ 1)$?

$$y = \begin{cases} 1\ or\ obese\ if\ p \geq 0.5 \\ 0\ or\ not\ obsese\ otherwise \end{cases}$$

# Logistic Regression as a Special Linear Layer



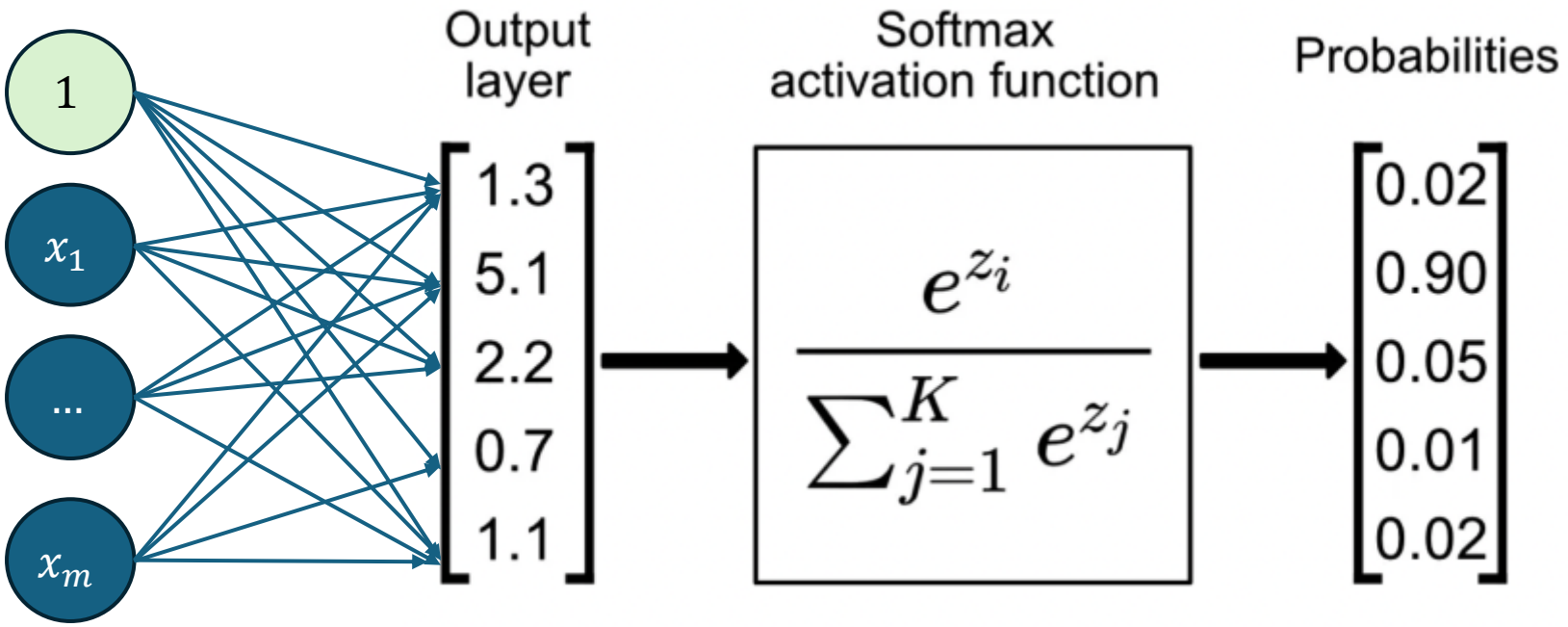
# Multiclass Logistic Regression / Softmax Classifier as a Special Linear Layer





$$\mathbf{Y} = g_1(\mathbf{W}\mathbf{X}^T)$$

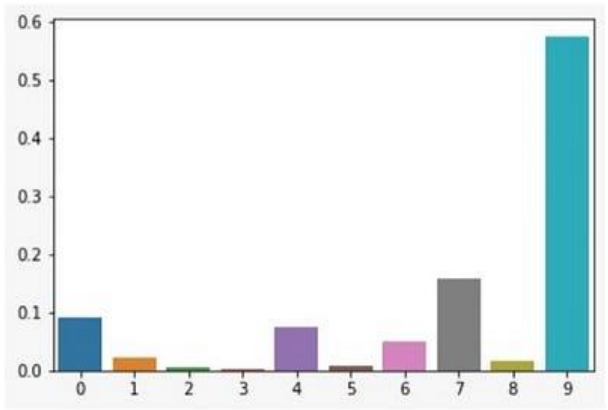
# Softmax Activation Function



# Softmax Activation Function

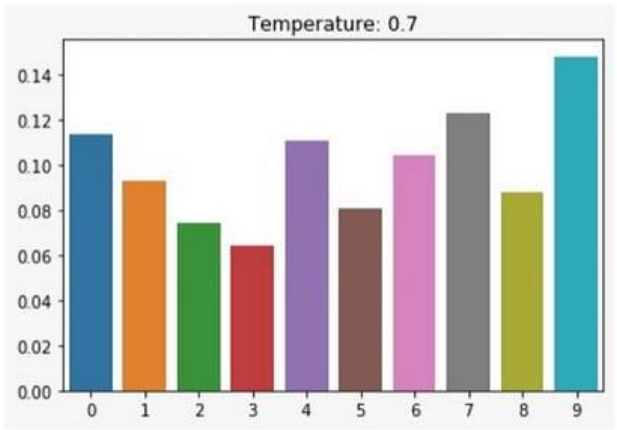
SOFTMAX WITHOUT TEMPERATURE (T=1)

$$\frac{e^{z_i}}{\sum_j e^{z_i}}$$

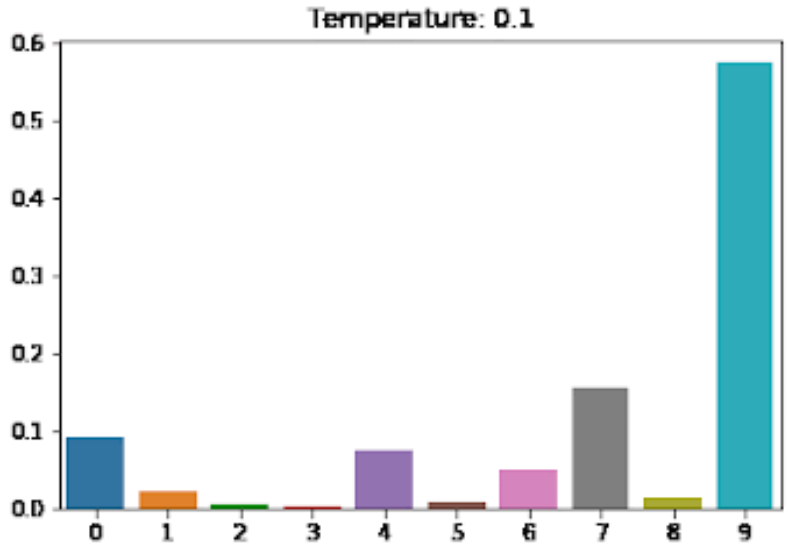



SOFTMAX WITH TEMPERATURE

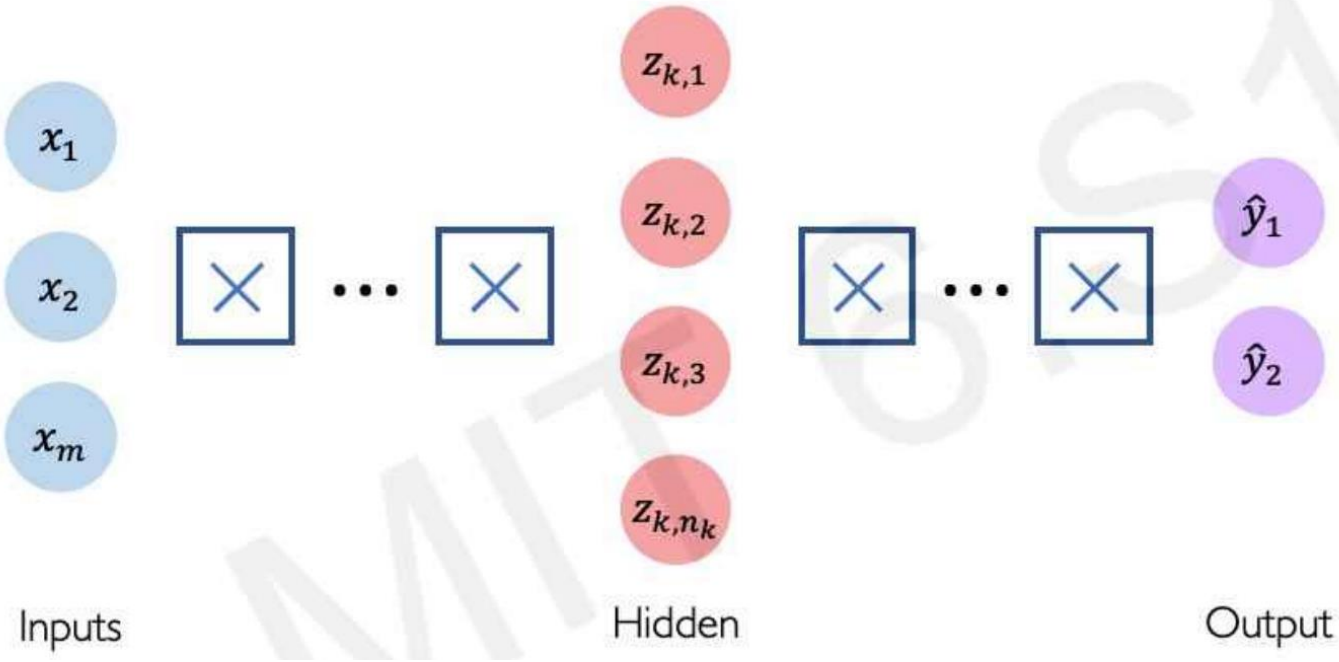
$$\frac{e^{z_i/T}}{\sum_j e^{z_i/T}}$$




LESS ENTROPY → INCREASE IN ENTROPY WITH INCREASE IN T → MORE ENTROPY

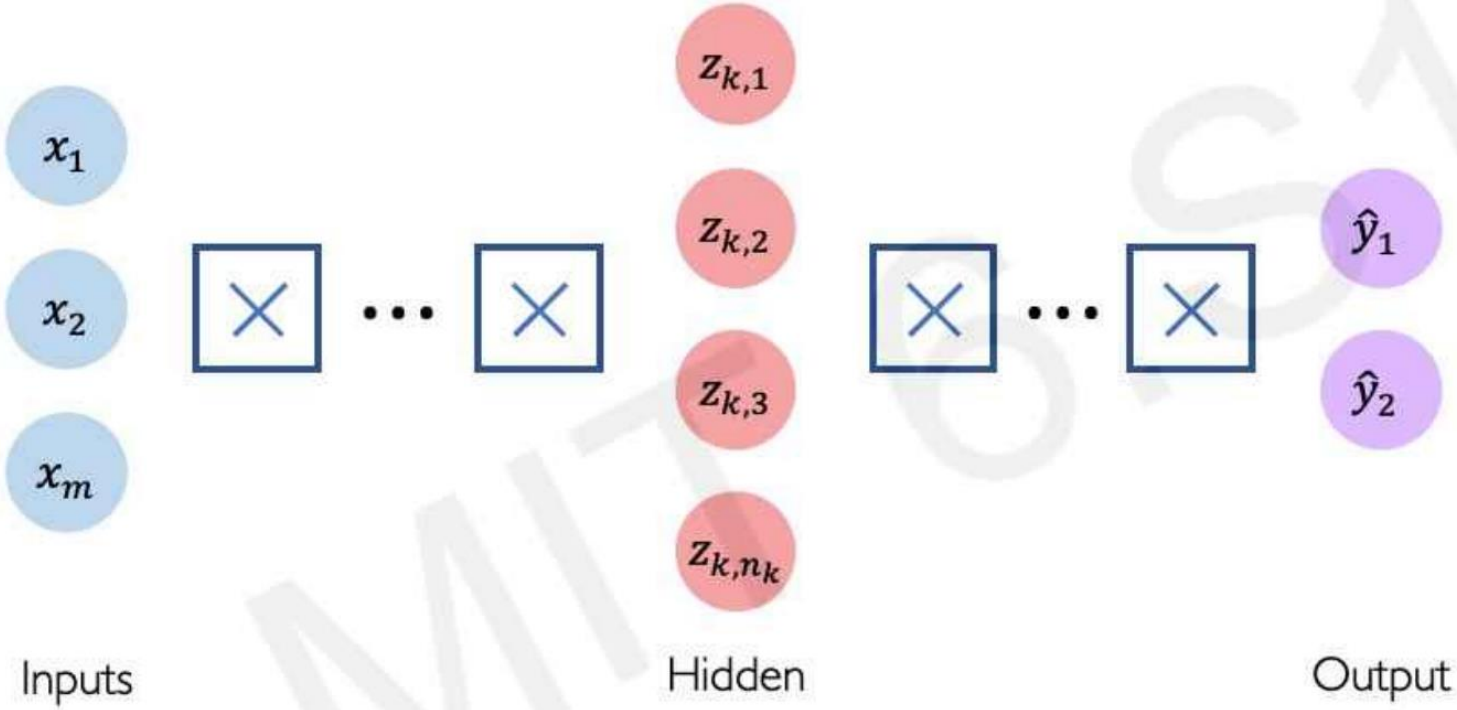


# Deep Neural Networks – Design




- How many input features - $m$
- How many hidden layers - $nlayers$
- How many hidden neurons in each layer ? – $[H_1, \ldots, H_l]$
- Each layer's activation function?
- Output activation – Sigmoid (binary classification), Softmax (multi-class classification), Identity (regression)

# Deep Neural Networks – Design in Practice

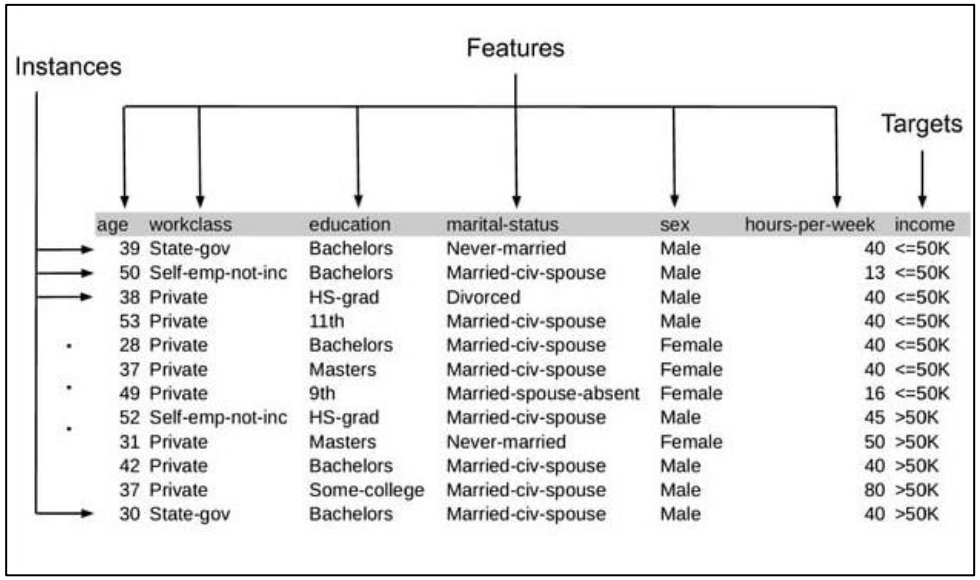



```tensorflow
import tensorflow as tf

model = tf.keras.Sequential([
  tf.keras.layers.Dense(n_1),
  tf.keras.layers.Dense(n_2),
  ⋮
  tf.keras.layers.Dense(2)
])
```

```pytorch
from torch import nn

model = nn.Sequential(
  nn.Linear(m, n1),
  nn.ReLU(),
⋮
  nn.ReLU(),
  nn.Linear(nK, 2)
)
```

# Pre-Processing

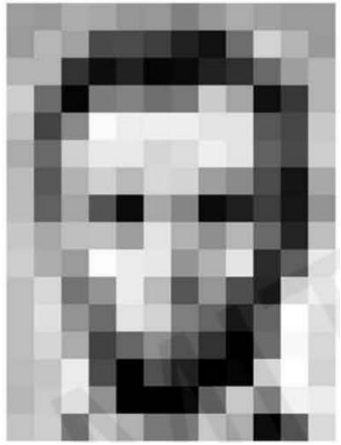
## Structured Data




| age | workclass | education | marital-status | sex | hours-per-week | income |
|---|---|---|---|---|---|---|
| 39 | State-gov | Bachelors | Never-married | Male | 40 | <=50K |
| 50 | Self-emp-not-inc | Bachelors | Married-civ-spouse | Male | 13 | <=50K |
| 38 | Private | HS-grad | Divorced | Male | 40 | <=50K |
| 53 | Private | 11th | Married-civ-spouse | Male | 40 | <=50K |
| 28 | Private | Bachelors | Married-civ-spouse | Female | 40 | <=50K |
| 37 | Private | Masters | Married-civ-spouse | Female | 40 | <=50K |
| 49 | Private | 9th | Married-spouse-absent | Female | 16 | <=50K |
| 52 | Self-emp-not-inc | HS-grad | Married-civ-spouse | Male | 45 | >50K |
| 31 | Private | Masters | Never-married | Female | 50 | >50K |
| 42 | Private | Bachelors | Married-civ-spouse | Male | 40 | >50K |
| 37 | Private | Some-college | Married-civ-spouse | Male | 80 | >50K |
| 30 | State-gov | Bachelors | Married-civ-spouse | Male | 40 | >50K |

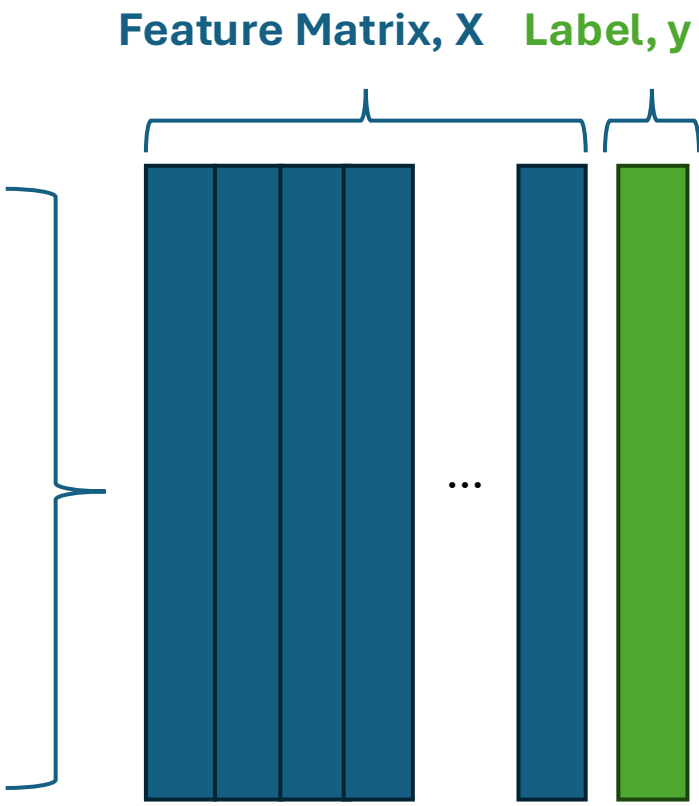
- Non-numerical inputs (i.e., categorical)
  - Nominal (no order)
  - Ordinal (order)
  - Binary
- Numeric inputs
  - Standardization
  - Min-max normalization

## Unstructured Data



*I go to school.*

- Flattening to vector
- Feature extraction/processing
- Numeric inputs
  - Standardization
  - Min-max normalization



# Non-Numerical Inputs (Categorical)

| Color |
|---|
| Red |
| Green |
| Blue |

**One-hot encoding**

| d1 | d2 | d3 |
|---|---|---|
| 1 | 0 | 0 |
| 0 | 1 | 0 |
| 0 | 0 | 1 |

| Original Encoding | Ordinal Encoding |
|---|---|
| Poor | 1 |
| Good | 2 |
| Very Good | 3 |
| Excellent | 4 |

| Color |
|---|
| Red |
| Green |
| Blue |

**Dummy encoding**

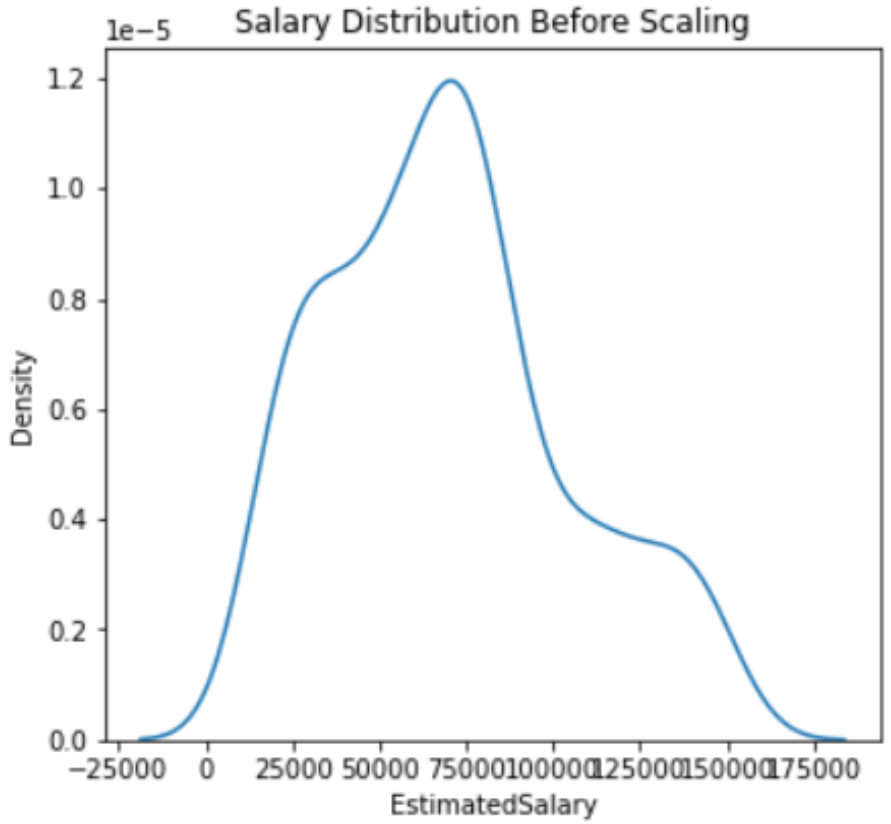
| d1 | d2 |
|---|---|
| 1 | 0 |
| 0 | 1 |
| 0 | 0 |

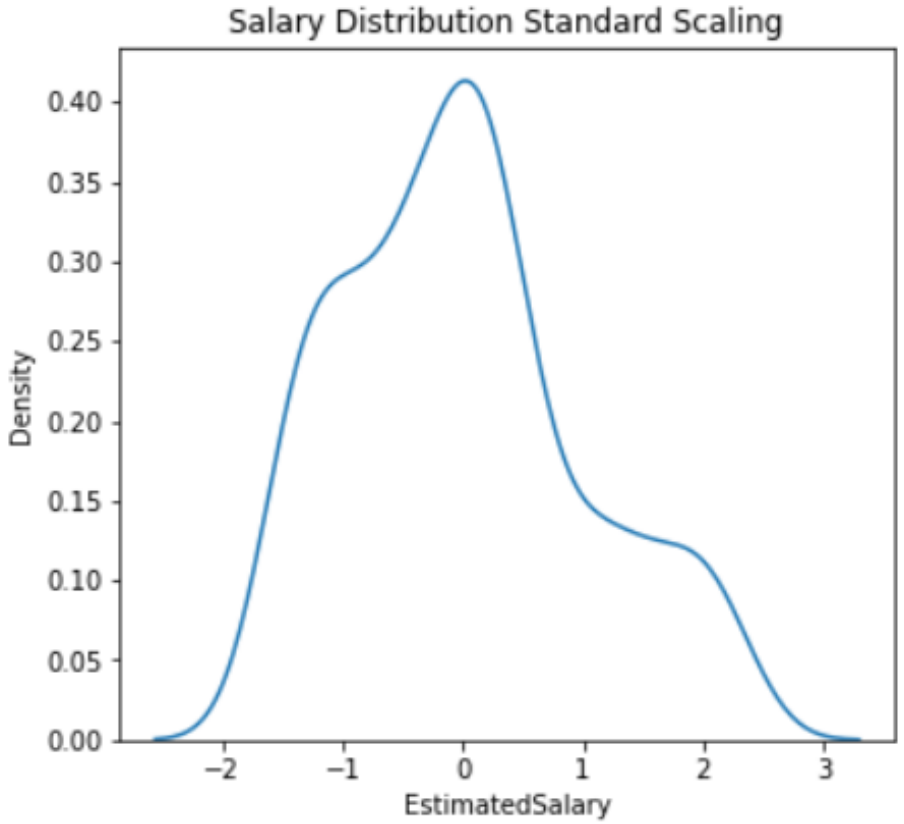
| Binary Encoding | Encoded |
|---|---|
| Yes | 1 |
| No | 0 |
| ... | |
| No | 0 |

# Numeric Inputs – Standardization

- **Concept:** Rescales the data so that it has a mean of 0 and a standard deviation of 1. It centers the data around zero.

$$z = \frac{x - \mu}{\sigma}$$

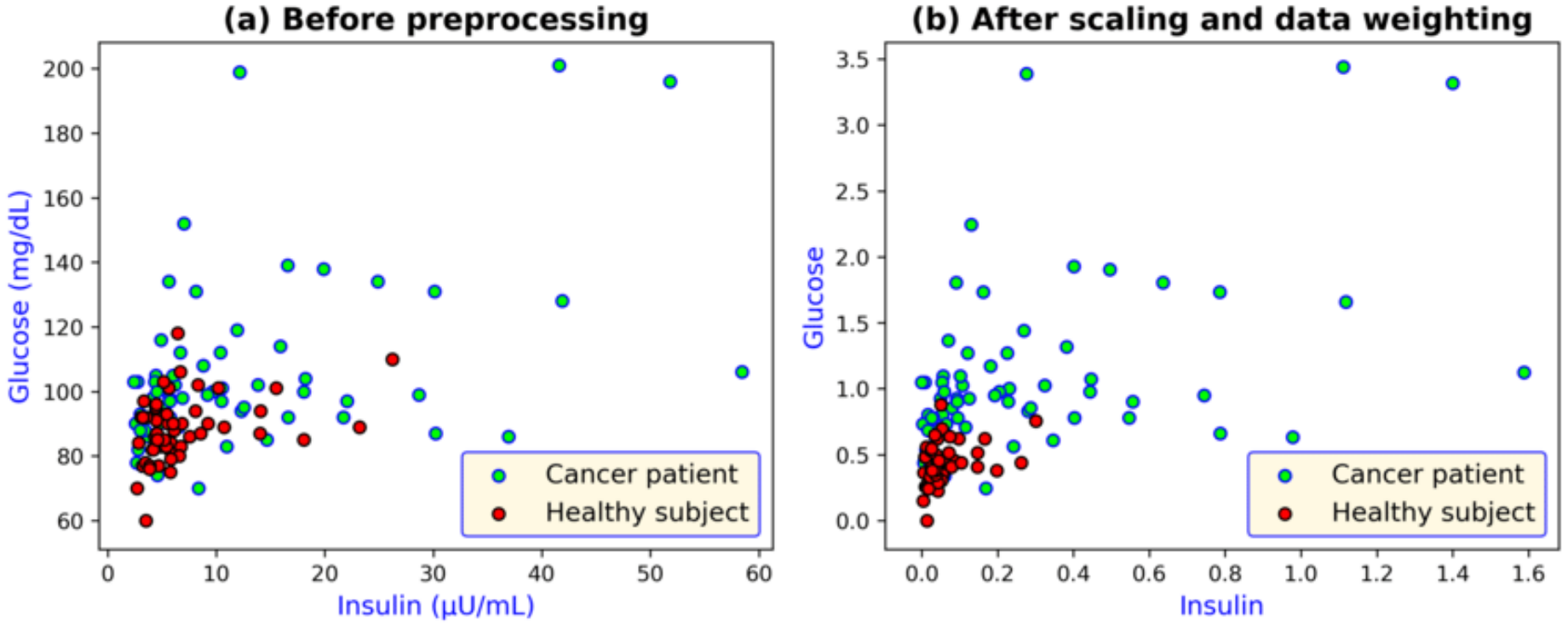


# Numeric Inputs – Min-max

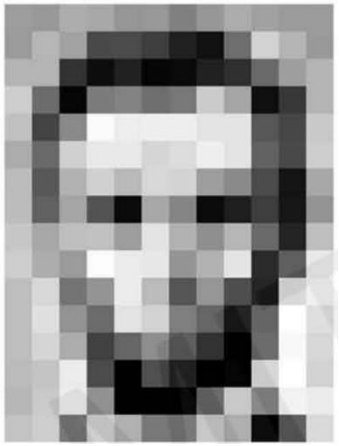
- **Concept:** Rescales the data so that all values fall exactly within a specific bounded range, usually between 0 and 1.

$$x_{norm} = \frac{x - x_{min}}{x_{max} - x_{min}}$$



# Feature Extraction

**Image**



$h$

$w$

**Flattening**

$$\begin{pmatrix} p_{1,1} \\ \dots \\ p_{h,w} \end{pmatrix}$$

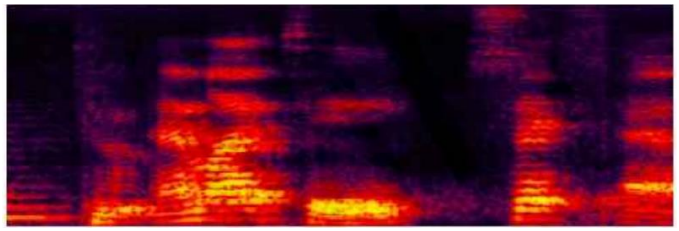
**Scaling**

$$\begin{pmatrix} p_{1,1} \\ \dots \\ p_{h,w} \end{pmatrix}$$

**Audio**

**Fourier analysis**

**Spectrogram Image**



*I go to school.*

**Text Feature Extractor**

$w_1$

$w_{vocab}$

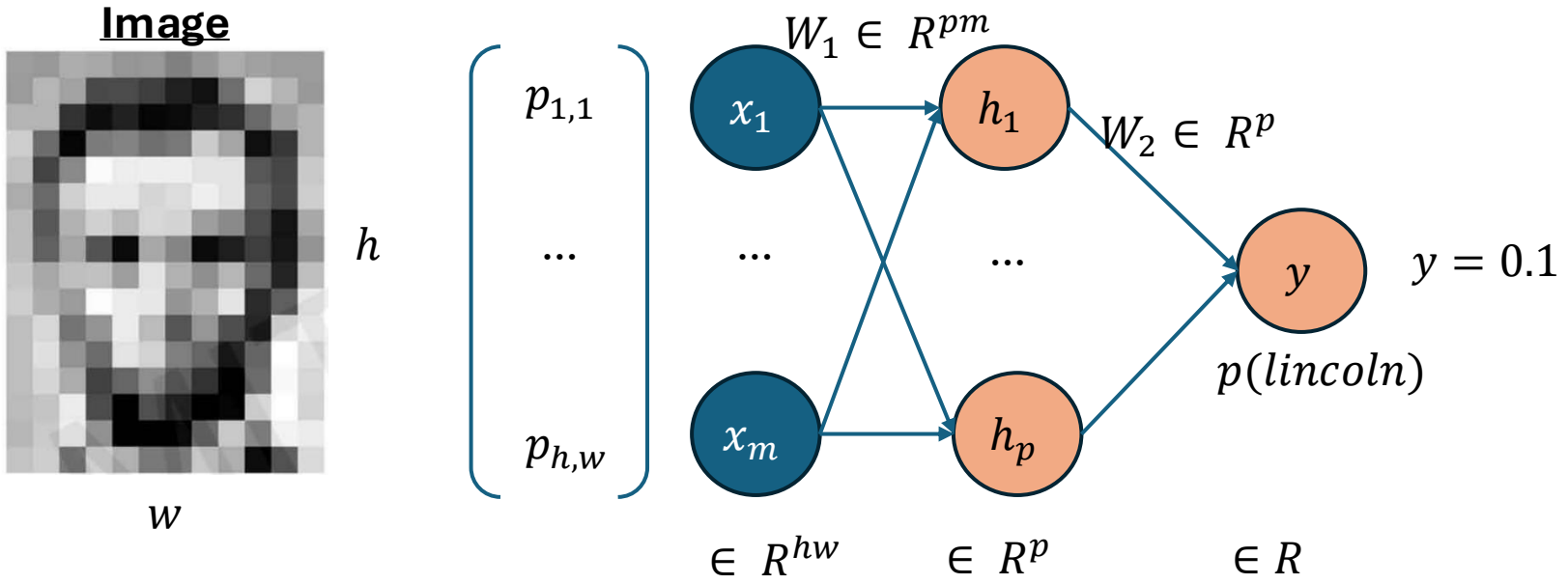
$$\begin{pmatrix} 5 \\ \dots \\ 0 \end{pmatrix}$$

**Term-Freq. Vector**

$$\begin{pmatrix} 0.1 \\ \dots \\ 0.0 \end{pmatrix}$$

**Normalized Term-Freq. Vector**

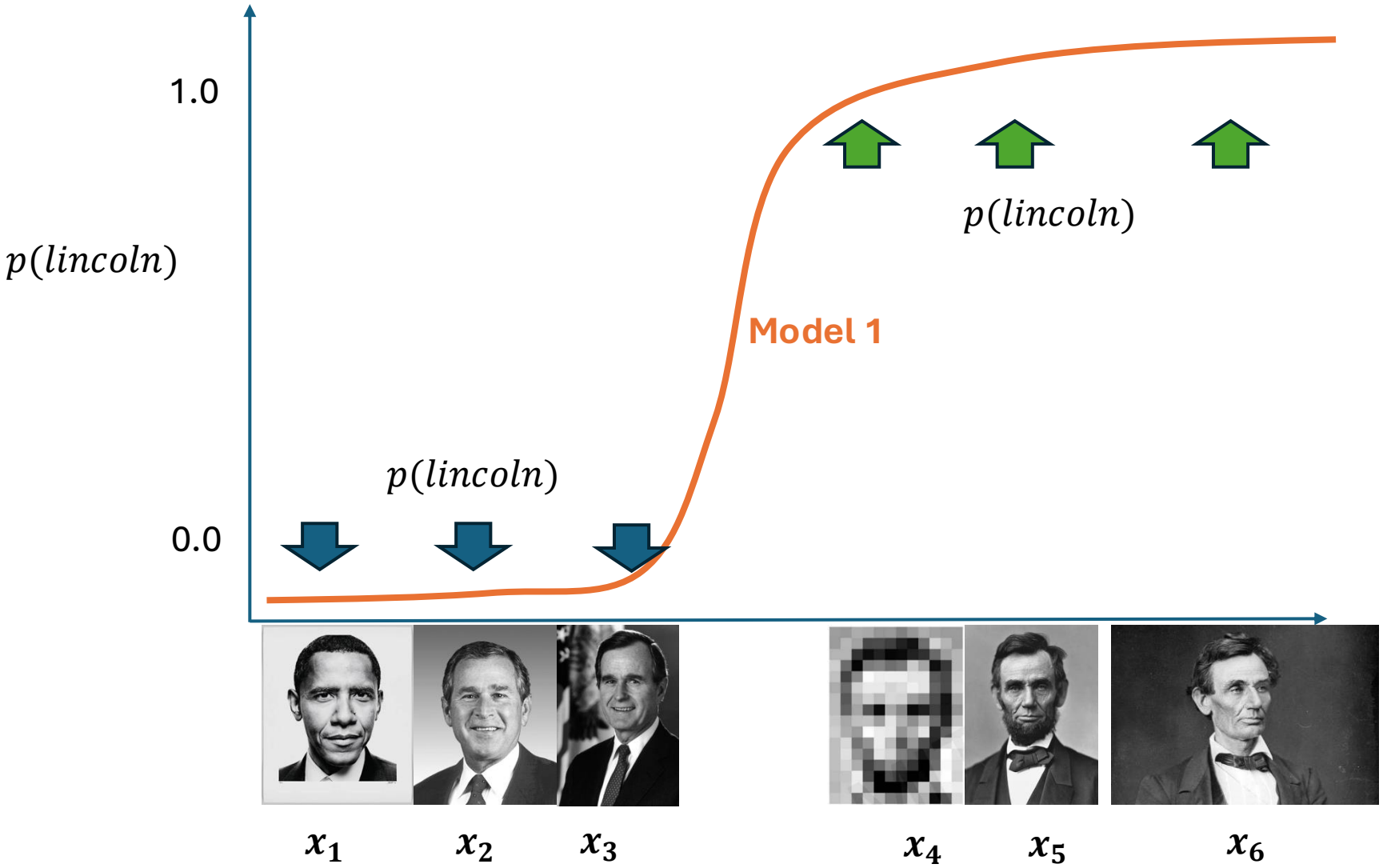
# Lincoln Classifier



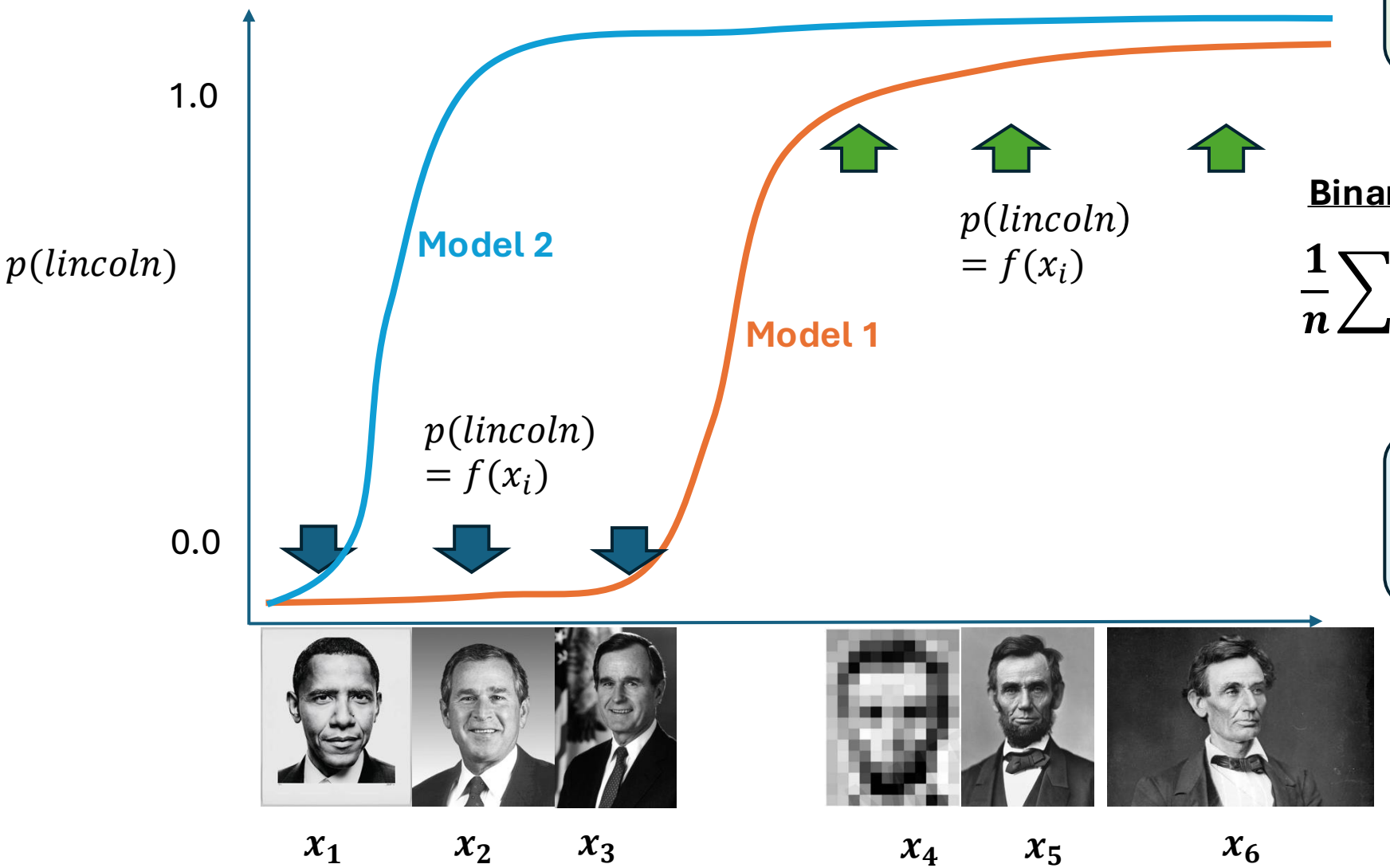

$$W_1 \in R^{pm} \sim N(0,1)$$

$$W_2 \in R^p \sim N(0,1)$$

# Lincoln Classifier



# Lincoln Classifier




$[x_4, x_5, x_6]$

$y_i \log(p_i)$

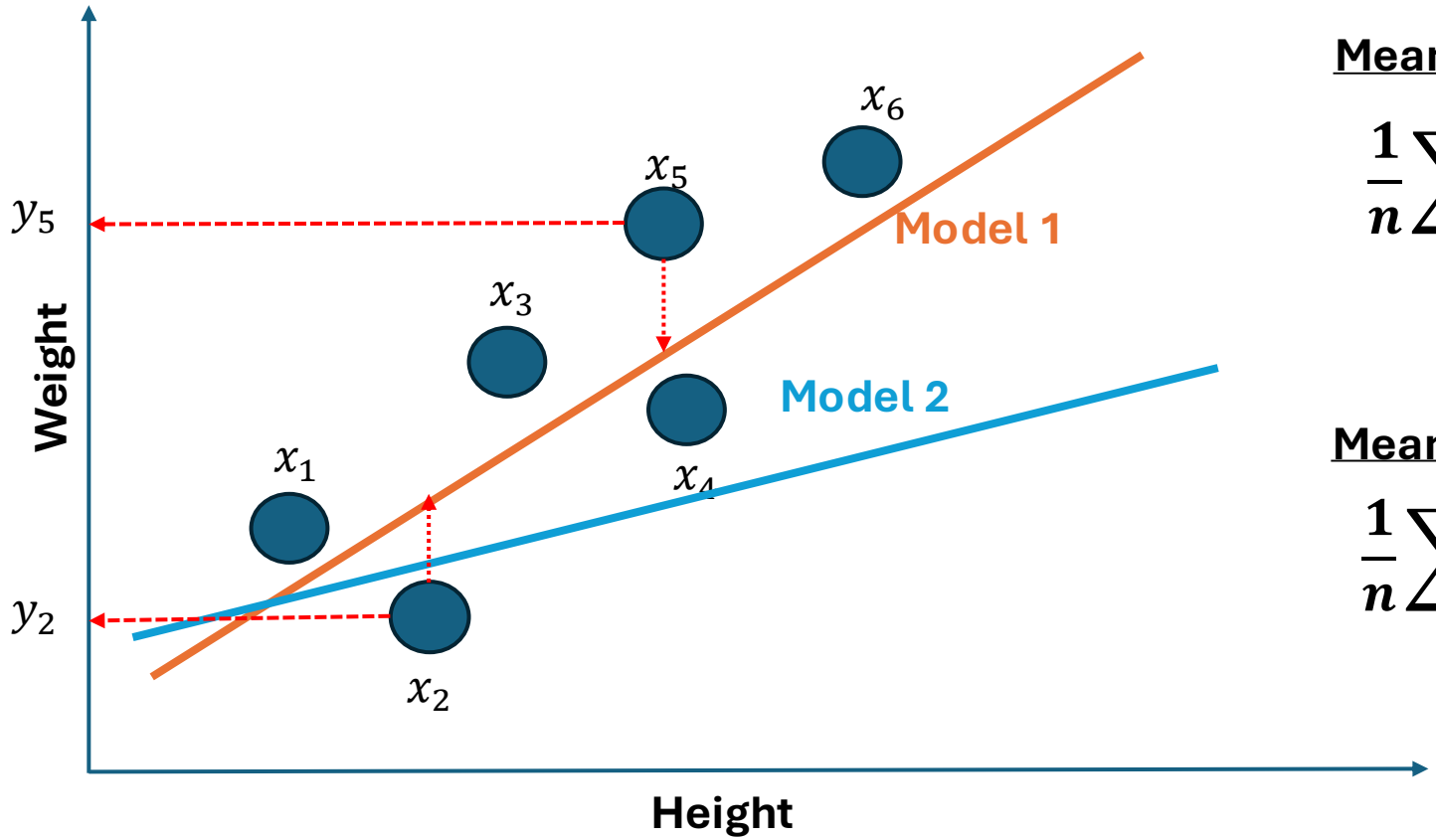
**Binary Cross-Entropy**

$$\frac{1}{n}\sum y_i \log(p_i) + (1 - y_i)\log(1 - p_i)$$

$[x_1, x_2, x_3]$

$(1 - y_i)\log(1 - p_i)$

# Lincoln Classifier




**Mean-Squared Error (MSE)**

$$\frac{1}{n}\sum\left(y_i - f(x_i)\right)^2$$

**Mean-Absolute Error (MSE)**

$$\frac{1}{n}\sum|y_i - f(x_i)|$$

# Loss / Cost Function

**Binary Cross-Entropy**

$$L = \frac{1}{n}\sum y_i \log(p_i) + (1 - y_i)\log(1 - p_i)$$

**Cross-Entropy**

$$L = \frac{1}{n}\sum y_i log(p_i)$$

*P for the true label for sample i*

*1 for true label for sample i*

**Mean-Squared Error (MSE)**

$$L = \frac{1}{n}\sum \left(y_i - f(x_i)\right)^2$$

**Mean-Absolute Error (MSE)**

$$L = \frac{1}{n}\sum |y_i - f(x_i)|$$

**Model Training as Loss Minimization**

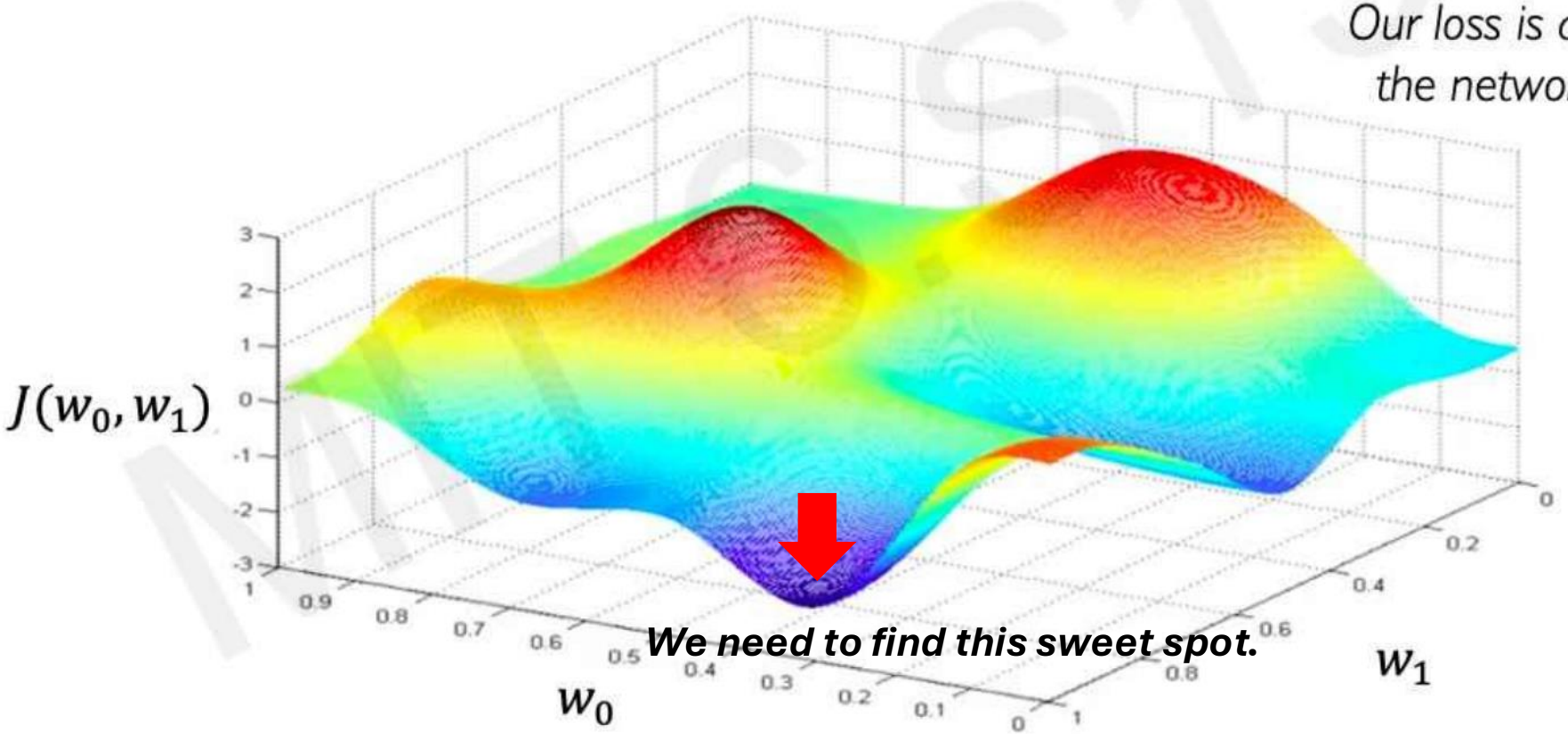
$$W^* = \underset{W}{\operatorname{argmin}} \frac{1}{n}\sum_{i=1}^{n} \mathcal{L}\left(f\left(x^{(i)}; W\right), y^{(i)}\right)$$

$$W^* = \underset{W}{\operatorname{argmin}} J(W)$$

# Loss Surface – Non-Convex

$$W^* = \underset{W}{\operatorname{argmin}} J(W)$$

Remember:
*Our loss is a function of the network weights!*

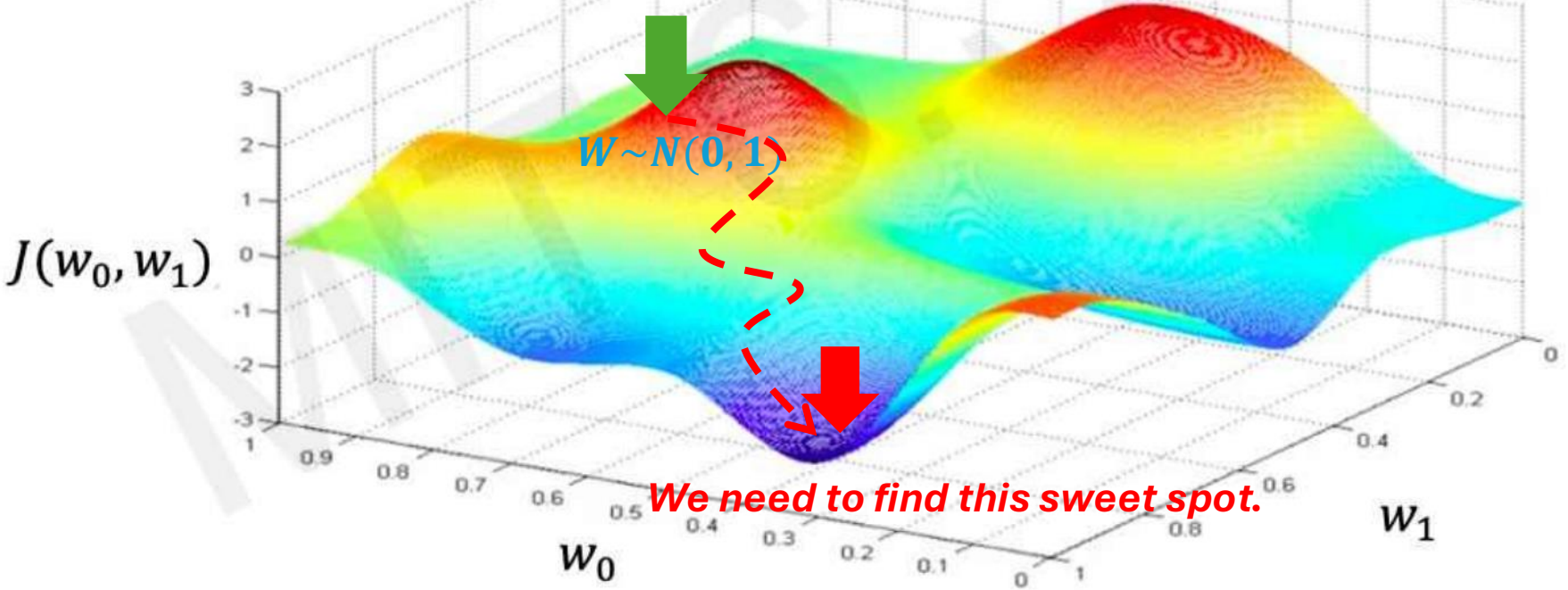



***We need to find this sweet spot.***

# Loss Surface – Non-Convex

$$W^* = \underset{W}{\operatorname{argmin}} J(W)$$

Remember:
*Our loss is a function of the network weights!*

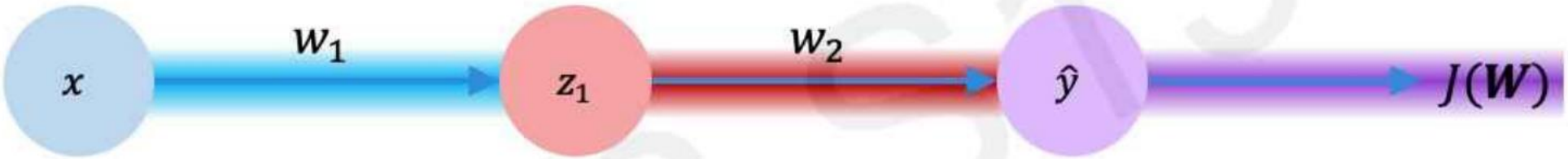


# Gradient Descent

**Algorithm**

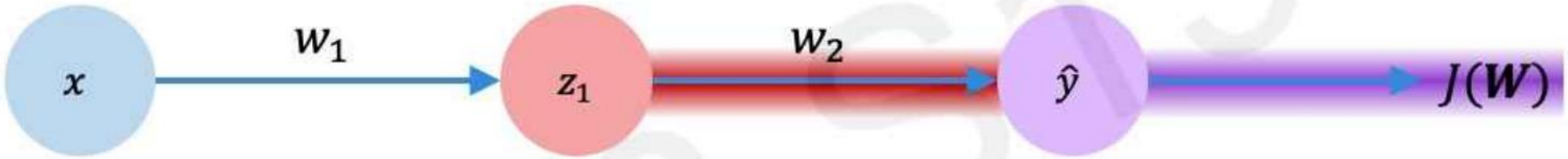
1. Initialize weights randomly $\sim \mathcal{N}(0, \sigma^2)$
2. Loop until convergence: $n_{epoch}$
3. Compute gradient, $\frac{\partial J(\boldsymbol{W})}{\partial \boldsymbol{W}}$ $Backpropagation$
4. Update weights, $\boldsymbol{W} \leftarrow \boldsymbol{W} - \eta \frac{\partial J(\boldsymbol{W})}{\partial \boldsymbol{W}}$ $Update\ or\ Step$ $learning\ rate$
5. Return weights $Model\ Weights$

# Computing Gradients: Backpropagation




$$\frac{\partial J(\boldsymbol{W})}{\partial w_1} = \frac{\partial J(\boldsymbol{W})}{\partial \hat{y}} * \frac{\partial \hat{y}}{\partial z_1} * \frac{\partial z_1}{\partial w_1}$$

# Computing Gradients: Backpropagation

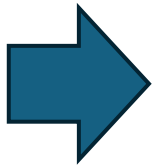



$$\frac{\partial J(\boldsymbol{W})}{\partial w_2} = \frac{\partial J(\boldsymbol{W})}{\partial \hat{y}} * \frac{\partial \hat{y}}{\partial w_2}$$

# Linear Regression as a Special Case

$$\boldsymbol{W}^* = \underset{\boldsymbol{W}}{\operatorname{argmin}} \frac{1}{n} \sum_{i=1}^{n} \mathcal{L}\left(f\left(x^{(i)}; \boldsymbol{W}\right), y^{(i)}\right)$$

$$\boldsymbol{W}^* = \underset{\boldsymbol{W}}{\operatorname{argmin}} J(\boldsymbol{W})$$



$$\frac{\partial J(W)}{\partial W} = 0$$

**Solve for $W$?**

$$W = (X^T X)^{-1} X^T y$$ ***Closed-form Solution***

# Closed-form Solution vs Non-closed-form Solution

$$2x + 3 = 7$$

$$x = \frac{7 - 3}{2} = 2$$

$$x = \cos(x)$$

**There is no simple algebraic expression for $x$.**

**Gradient Descent (iterative, approximate)**

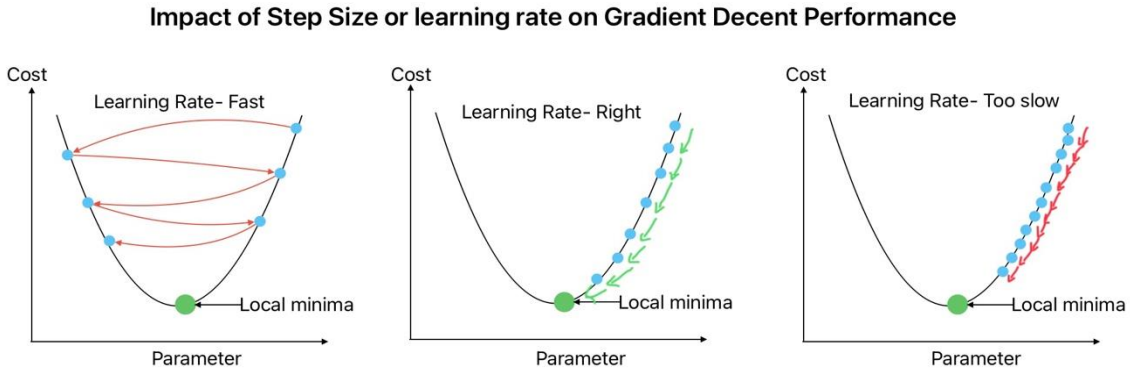
# Training Neural Networks is Difficult

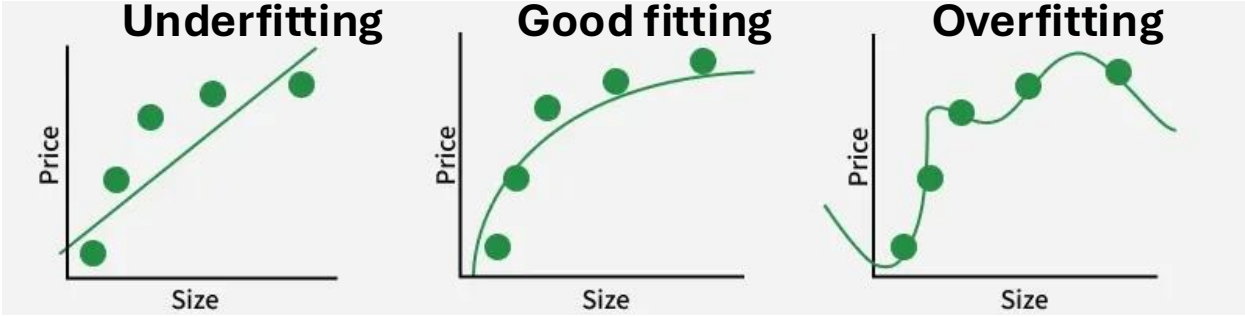
- Hyper-parameters
  - Optimizer-related
    - Learning Rate
    - Epochs
    - Etc.

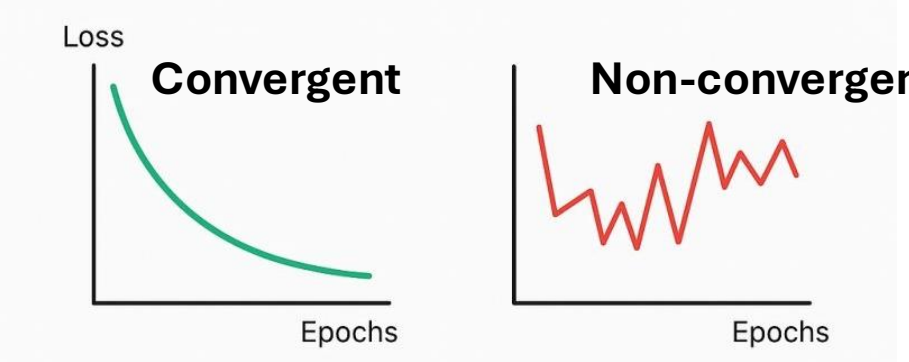
    ***Training stability***
  - Model-related
    - $n$ layers
    - number of neurons
    - loss functions
    - Etc.

    ***Model Generalization***
  - Pre-processing-related
    - Standardization
    - Min-max
    - Etc.

    ***Model Convergence***
  - Others



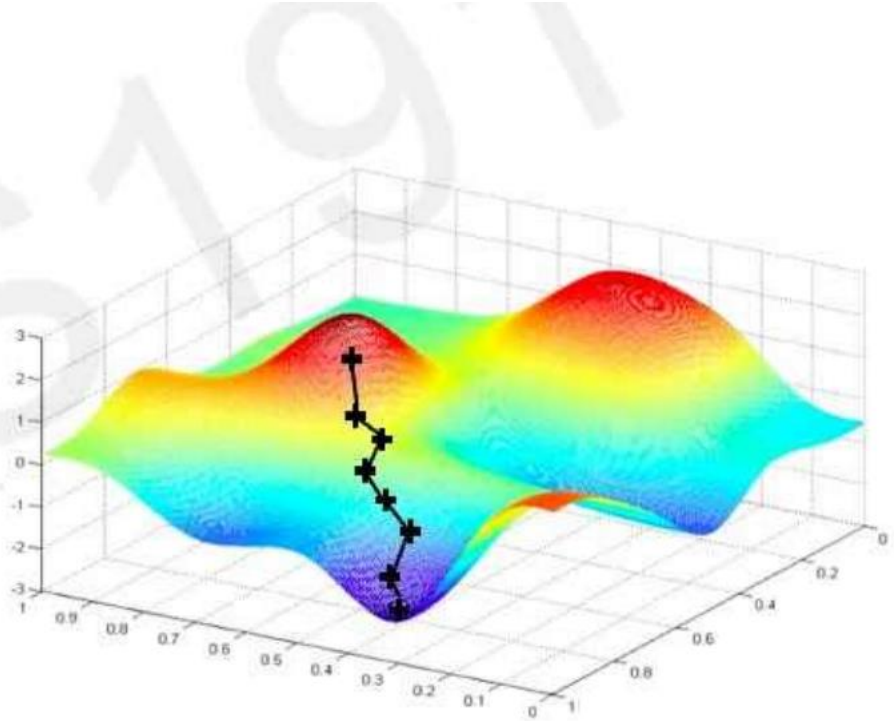
# Adaptive Learning Rate

| Algorithm | TF Implementation | Torch Implementation | Reference |
| --- | --- | --- | --- |
| • SGD | `tf.keras.optimizers.SGD` | `torch.optim.SGD` | Kiefer & Wolfowitz, 1952. |
| • Adam | `tf.keras.optimizers.Adam` | `torch.optim.Adam` | Kingma et al., 2014. |
| • Adadelta | `tf.keras.optimizers.Adadelta` | `torch.optim.Adadelta` | Zeiler et al., 2012. |
| • Adagrad | `tf.keras.optimizers.Adagrad` | `torch.optim.Adagrad` | Duchi et al., 2011. |
| • RMSProp | `tf.keras.optimizers.RMSProp` | `torch.optim.RMSProp` | |

# Vanilla Gradient Descent

**Algorithm**

1. Initialize weights randomly $\sim \mathcal{N}(0, \sigma^2)$
2. Loop until convergence:
3. Compute gradient, $\frac{\partial J(\boldsymbol{W})}{\partial \boldsymbol{W}}$
4. Update weights, $\boldsymbol{W} \leftarrow \boldsymbol{W} - \eta \frac{\partial J(\boldsymbol{W})}{\partial \boldsymbol{W}}$
5. Return weights



$$J(w) = \sum_{i=1}^{n} L_i$$

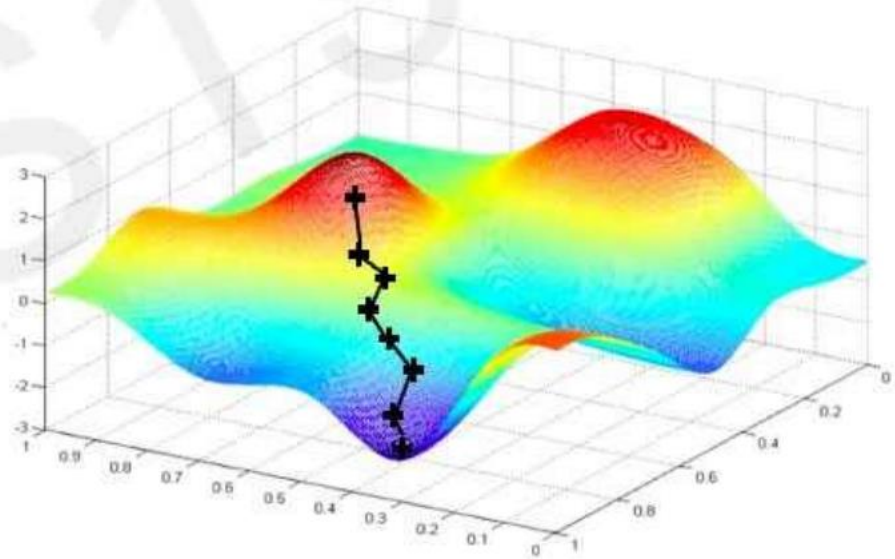
$O(n)$ complexity

What if 1M samples ?

# Stochastic Gradient Descent

**Algorithm**

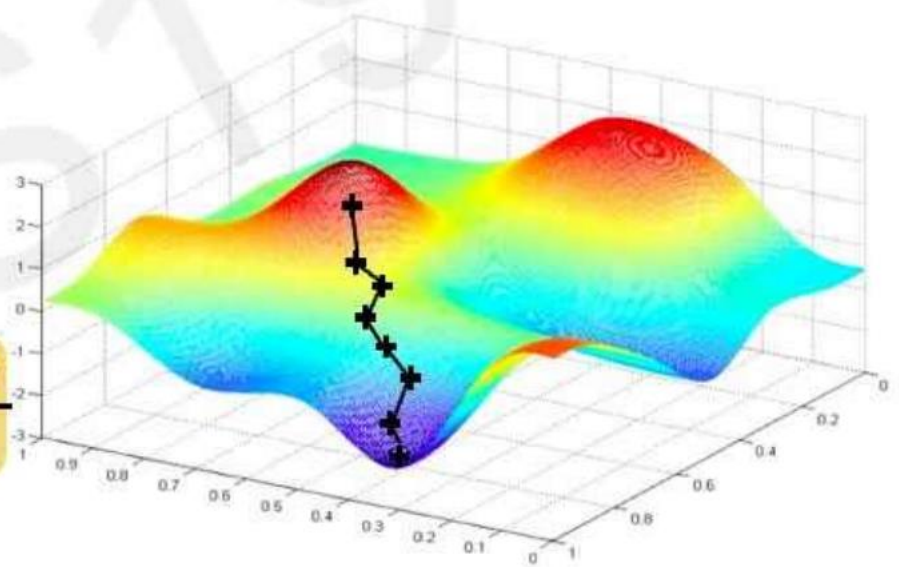
1. Initialize weights randomly $\sim \mathcal{N}(0, \sigma^2)$
2. Loop until convergence:
3. Pick single data point $\boldsymbol{i}$
4. Compute gradient, $\frac{\partial J_i(\boldsymbol{W})}{\partial \boldsymbol{W}}$
5. Update weights, $\boldsymbol{W} \leftarrow \boldsymbol{W} - \eta \frac{\partial J(\boldsymbol{W})}{\partial \boldsymbol{W}}$
6. Return weights



# Stochastic Gradient Descent

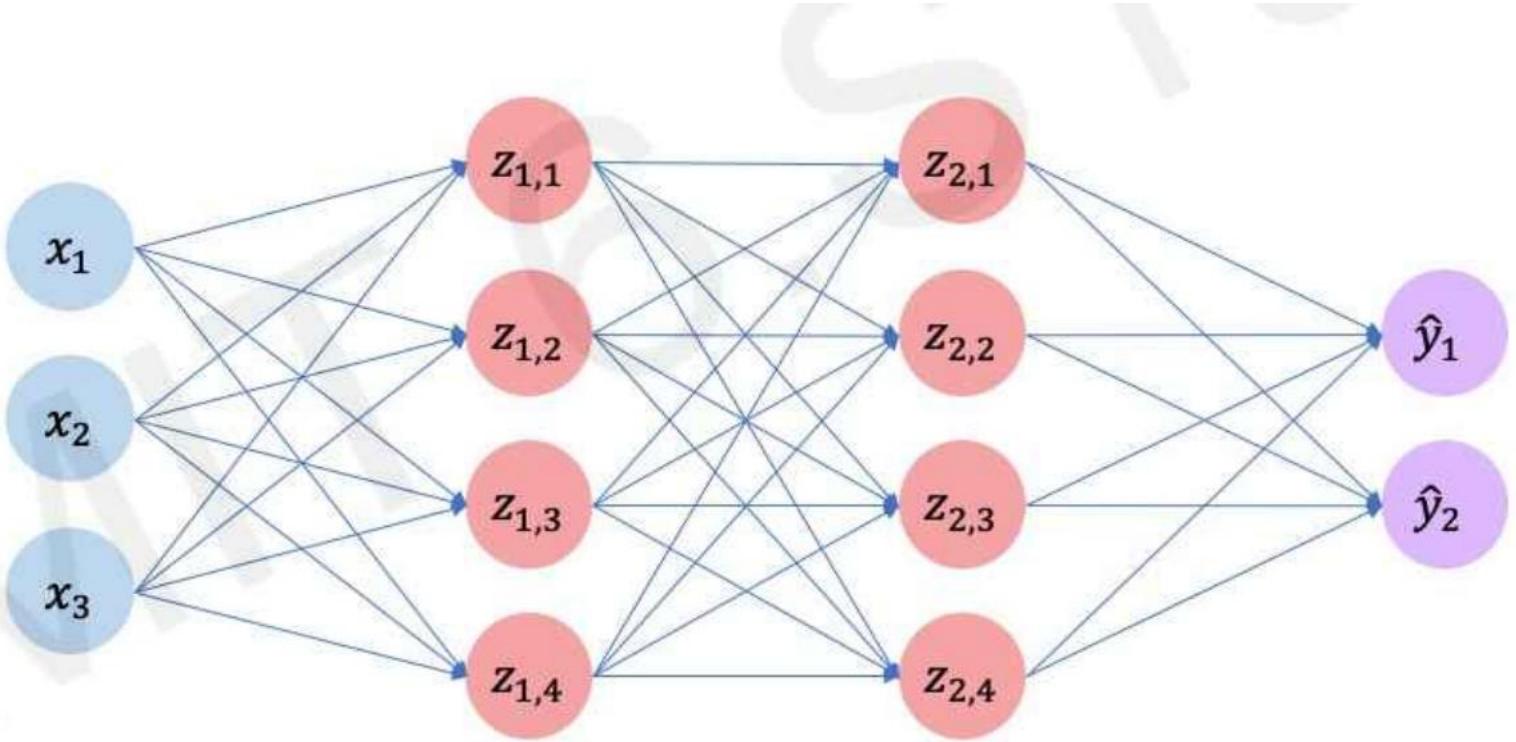
**Algorithm**

1. Initialize weights randomly $\sim \mathcal{N}(0, \sigma^2)$
2. Loop until convergence:
3. Pick batch of $B$ data points
4. Compute gradient, $\frac{\partial J(W)}{\partial W} = \frac{1}{B}\sum_{k=1}^{B} \frac{\partial J_k(W)}{\partial W}$
5. Update weights, $W \leftarrow W - \eta \frac{\partial J(W)}{\partial W}$
6. Return weights



Fast to compute and a much better estimate of the true gradient!

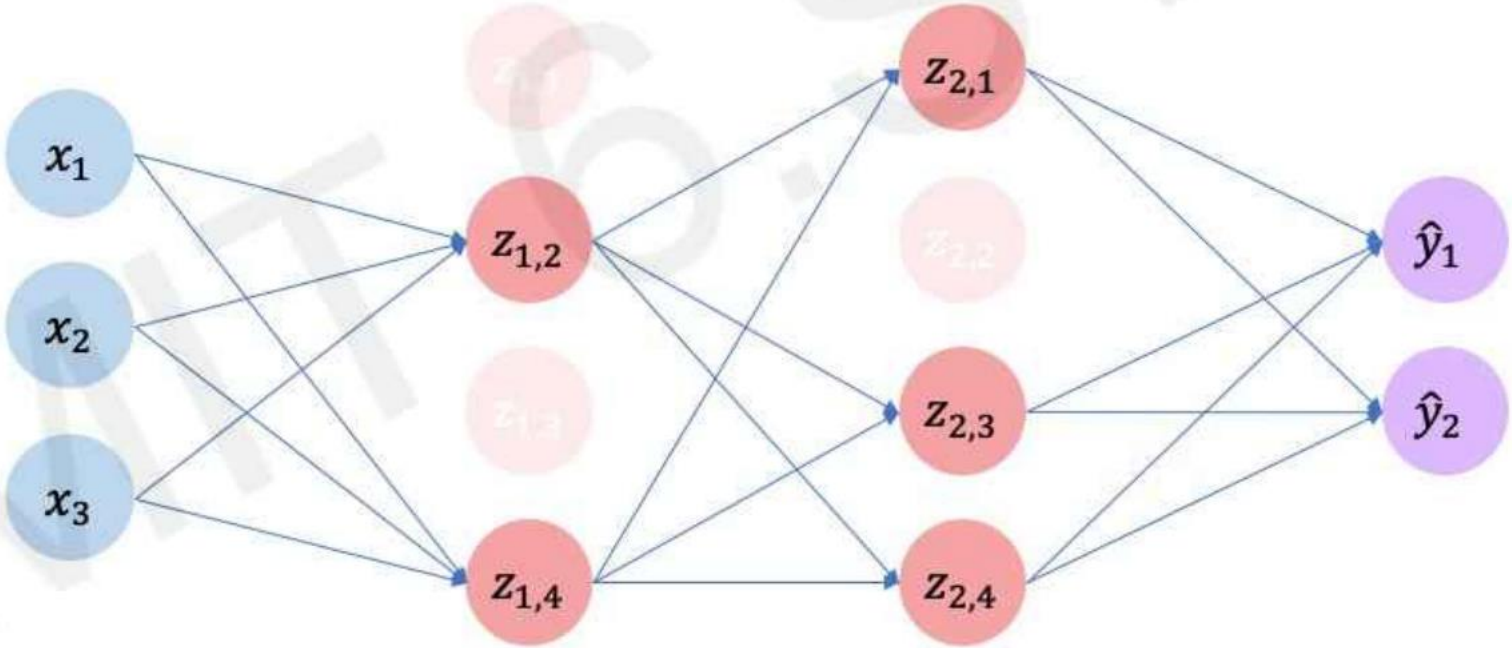
# Regularization



# Regularization - Dropout

- During training, randomly set some activations to 0
  - Typically 'drop' 50% of activations in layer
  - Forces network to not rely on any 1 node

```
tf.keras.layers.Dropout(p=0.5)
torch.nn.Dropout(p=0.5)
```



# Regularization - Dropout

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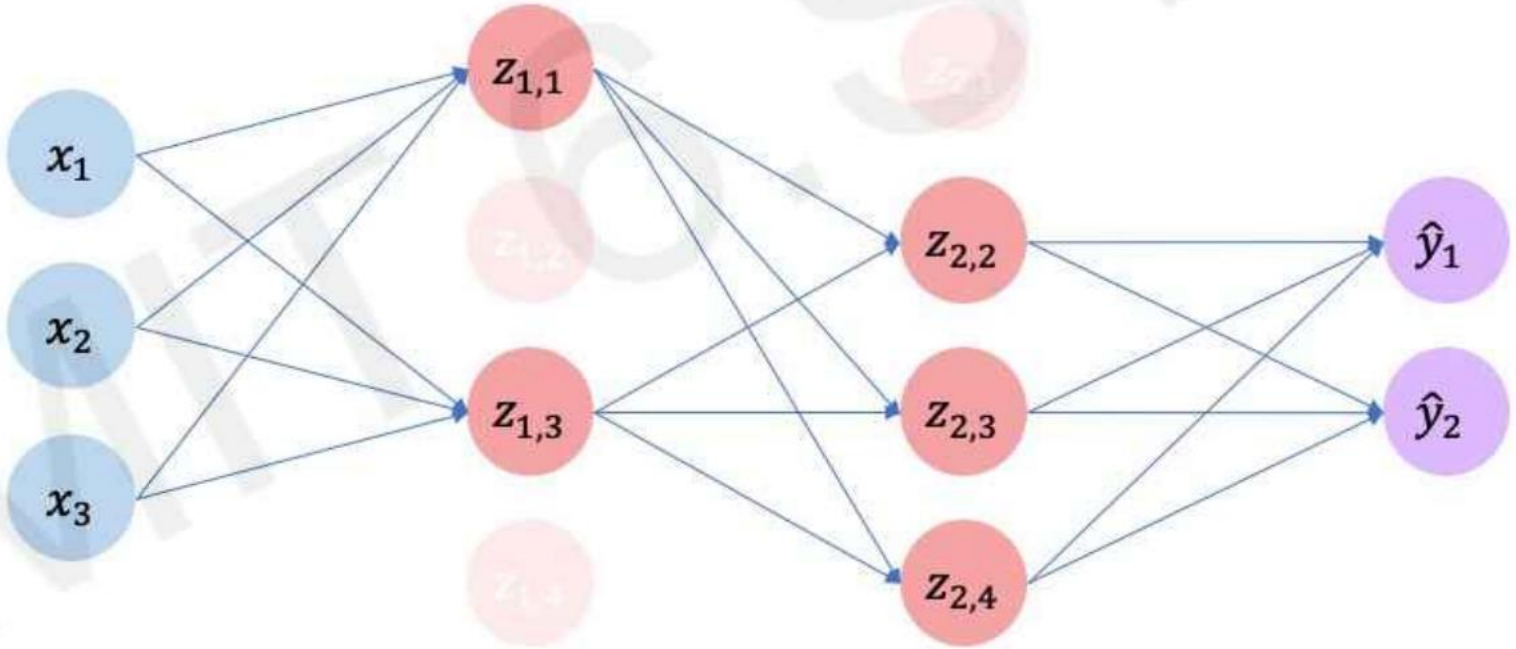


# Early Stopping

- Stop training before we have a chance to overfit

# Regularization

## L1 Regularization

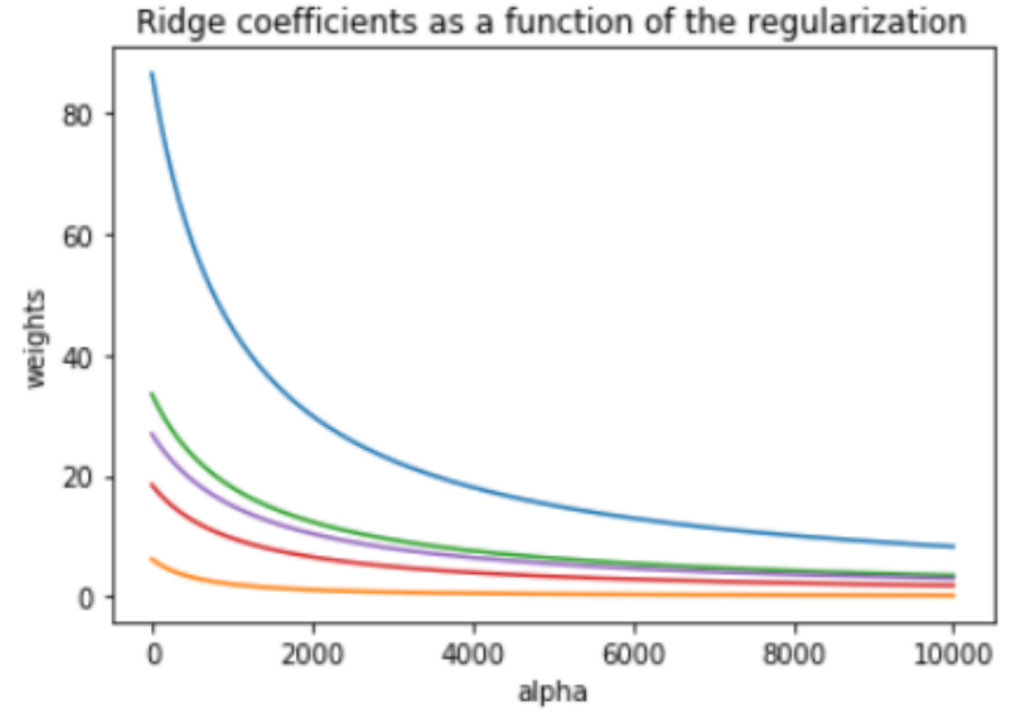
$$\text{Modified loss function} = \text{Loss function} + \lambda \sum_{i=1}^{n} |W_i|$$

***Regularization Factor (sometimes, $\alpha$)***

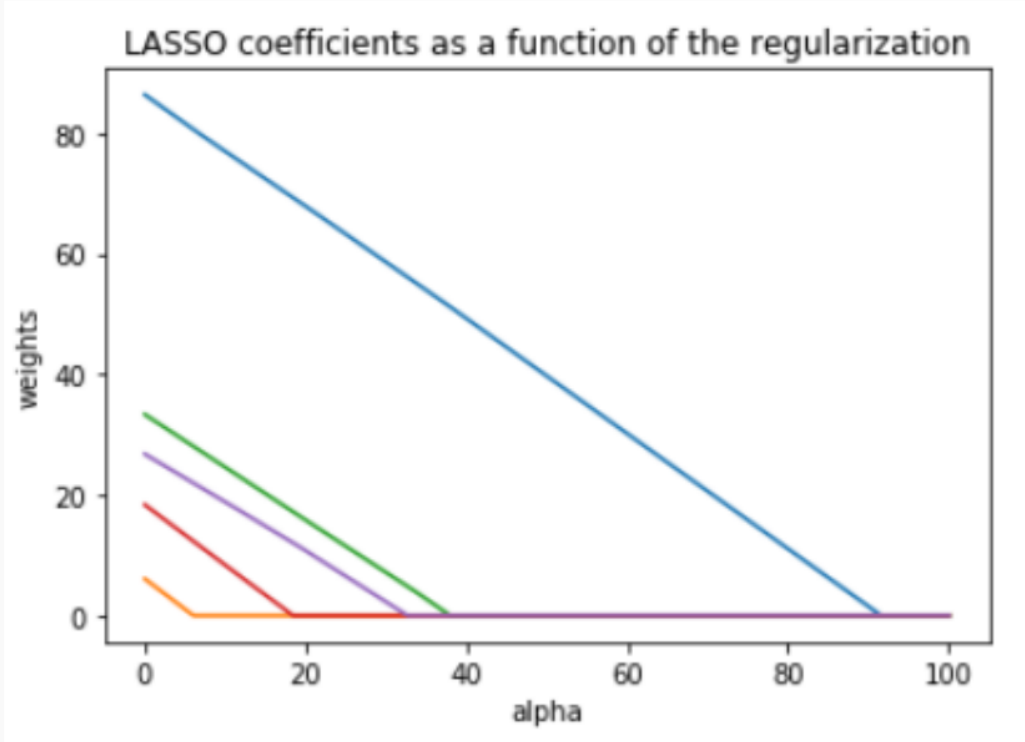
## L2 Regularization

$$\text{Modified loss function} = \text{Loss function} + \lambda \sum_{i=1}^{n} W_i^2$$

# Linear Regression as a Special Case – Ridge vs Lasso

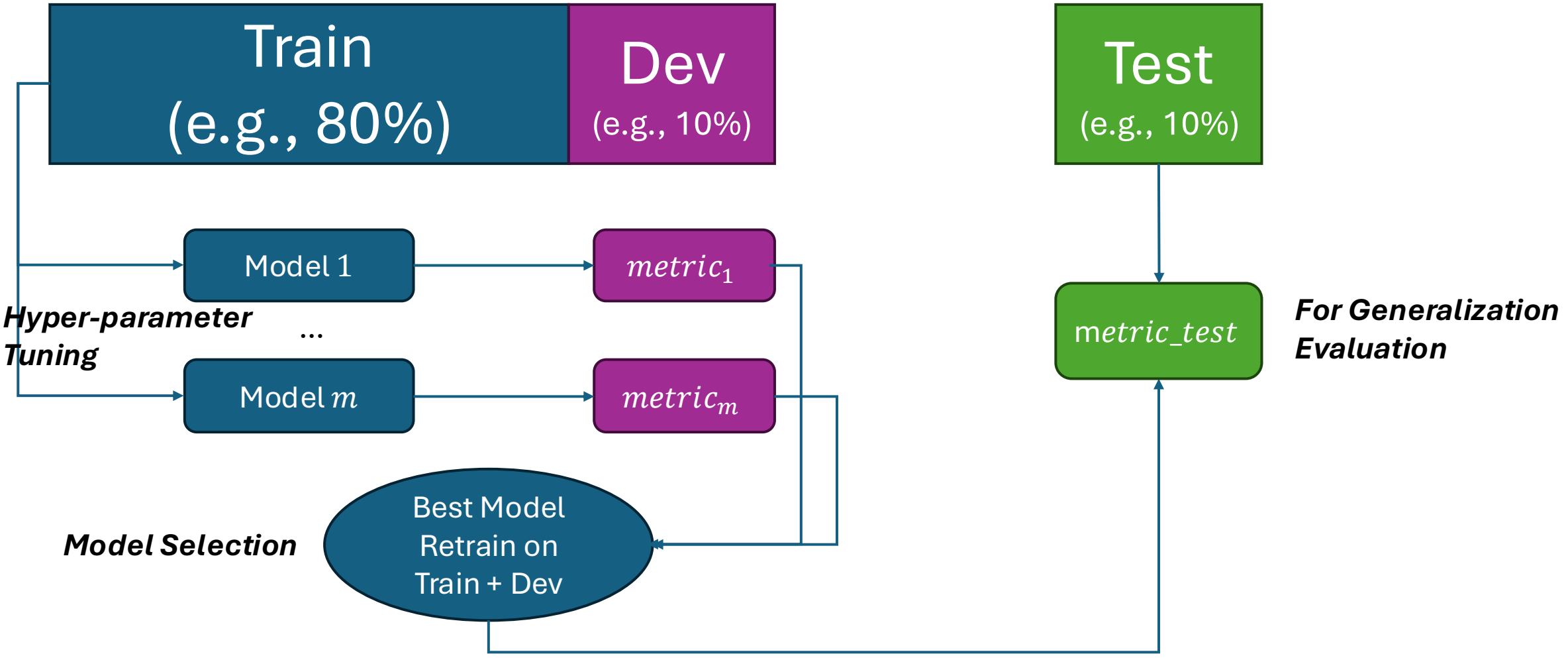



The values of the coefficients decrease as lambda increases, but they are not nullified.

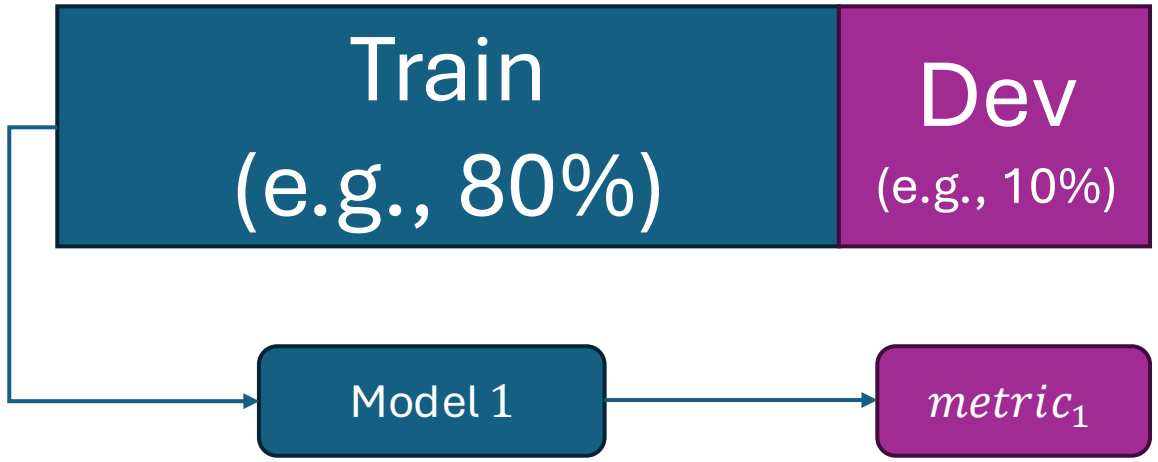



The values of the coefficients decrease as lambda increases, and are nullified fast.

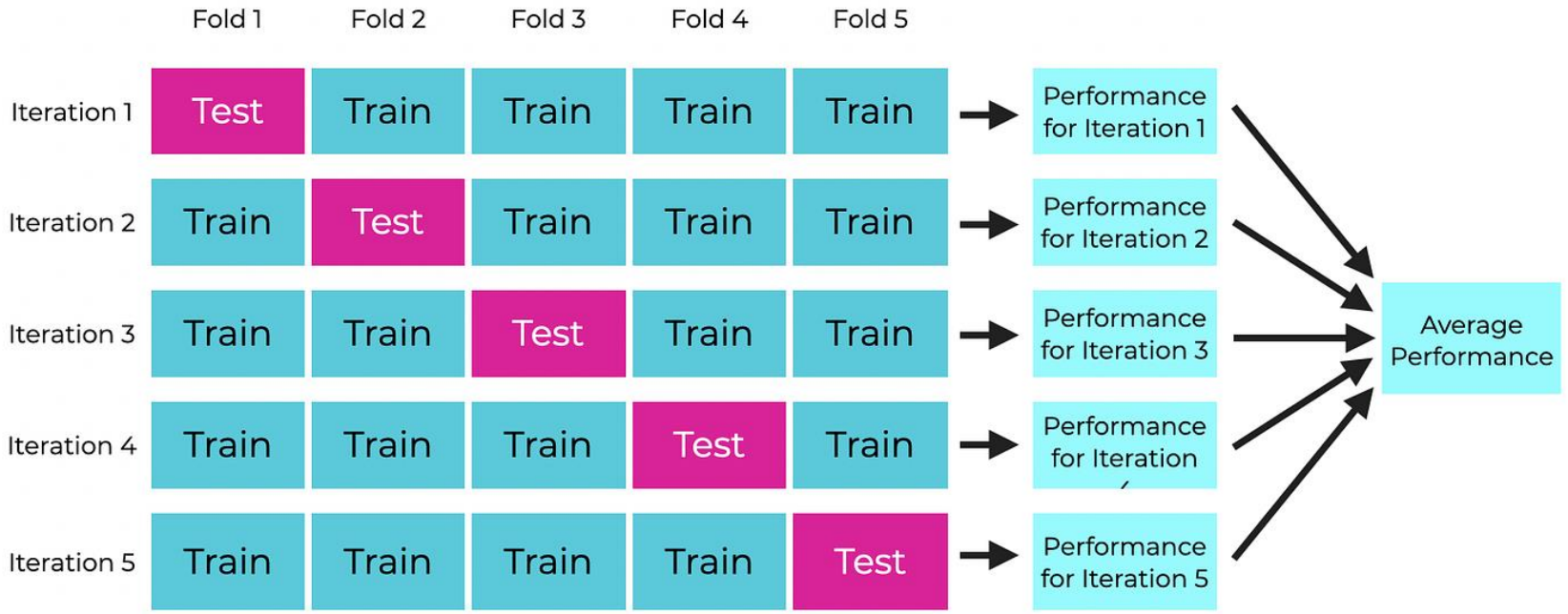
# Train-Development/Validation-Test Split



# K-Fold Cross-Validation




Common for Deep Learning




Common for Machine Learning

# Recap - Loss / Cost Function

**Binary Cross-Entropy**

$$L = \frac{1}{n}\sum y_i \log(p_i) + (1 - y_i)\log(1 - p_i)$$

**Cross-Entropy**

$$L = \frac{1}{n}\sum y_i log(p_i)$$

*P for the true label for sample i*

*1 for true label for sample i*

**Mean-Squared Error (MSE)**

$$L = \frac{1}{n}\sum \left(y_i - f(x_i)\right)^2$$

**Mean-Absolute Error (MSE)**

$$L = \frac{1}{n}\sum |y_i - f(x_i)|$$

These are just a proxy to a human-readable performance.

**Model Training as Loss Minimization**

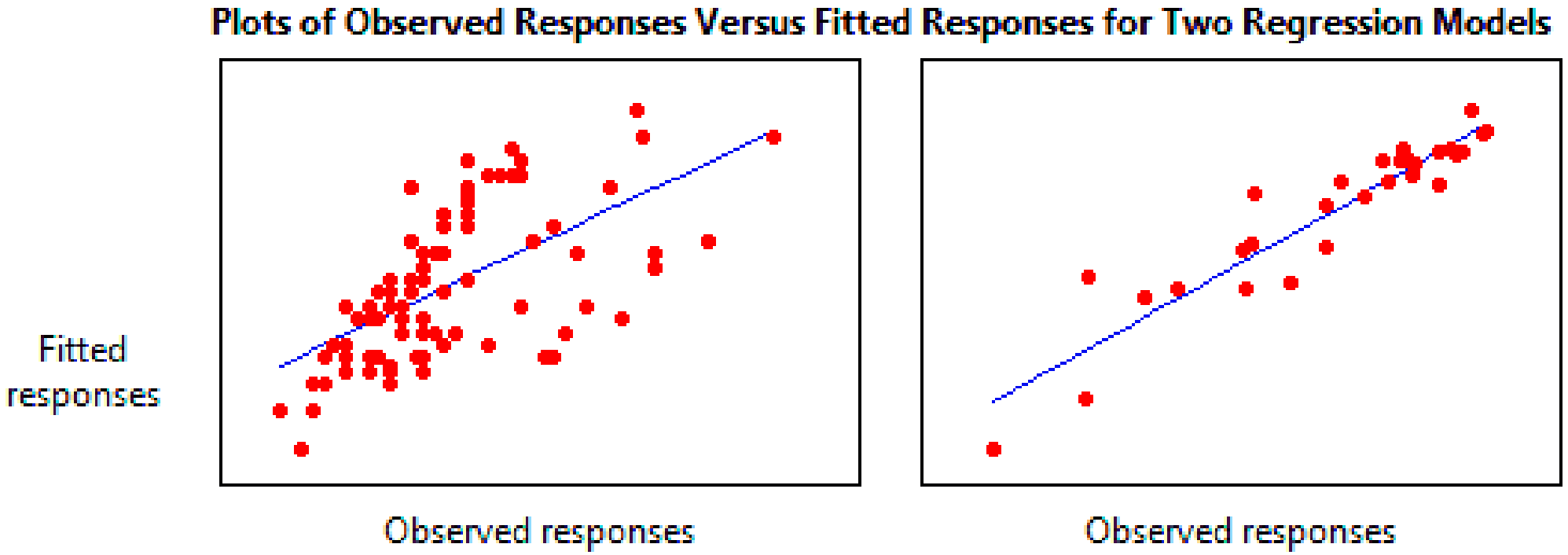
$$W^* = \underset{W}{\operatorname{argmin}} \frac{1}{n}\sum_{i=1}^{n} \mathcal{L}\left(f\left(x^{(i)}; W\right), y^{(i)}\right)$$

$$W^* = \underset{W}{\operatorname{argmin}} J(W)$$

# R-Squared for Regression Problems – [0,1]

It essentially acts as a baseline evaluation metric to score how well your predictive model fits your observed data.

$$R^2 = 1 - \frac{\Sigma_i^N (y_i - \hat{y}_i)^2}{\Sigma_i^N (y_i - \bar{y})^2}$$



# Confusion matrix for Binary Classification

When a classification algorithm (like logistic regression) is used, the results can be summarize in a ($k$ x $k$) table as such:

| | Predicted no AHD ($\hat{Y} = 0$) | Predicted AHD ($\hat{Y} = 1$) |
|---|---|---|
| Truly no AHD ($Y = 0$) | 110 | 54 |
| Truly AHD ($Y = 1$) | 53 | 86 |

The table above was a classification based on a logistic regression model to predict AHD based on "3" predictors: $X_1$ = Age, $X_2$ = Sex, and $X_3$ = interaction between Age and Sex.

What are the false positive and false negative rates for this classifier?

# Precision, Recall, and F1-Score

**Key:** **tp** = True Positive, **tn** = True Negative, **fp** = False Positive, **fn** = False Negative

| Metric Name | Metric Forumla | Code | When to use |
|---|---|---|---|
| Accuracy | $\textbf{Accuracy} = \frac{tp + tn}{tp + tn + fp + fn}$ | `tf.keras.metrics.Accuracy()` or `sklearn.metrics.accuracy_score()` | Default metric for classification problems. Not the best for imbalanced classes. |
| Precision | $\textbf{Precision} = \frac{tp}{tp + fp}$ | `tf.keras.metrics.Precision()` or `sklearn.metrics.precision_score()` | Higher precision leads to less false positives. |
| Recall | $\textbf{Recall} = \frac{tp}{tp + fn}$ | `tf.keras.metrics.Recall()` or `sklearn.metrics.recall_score()` | Higher recall leads to less false negatives. |
| F1-score | $\textbf{F1-score} = 2 \cdot \frac{\text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}}$ | `sklearn.metrics.f1_score()` | Combination of precision and recall, usually a good overall metric for a classification model. |
| Confusion matrix | NA | `Custom function` or `sklearn.metrics.confusion_matrix()` | When comparing predictions to truth labels to see where model gets confused. Can be hard to use with large numbers of classes. |

# Other Confusion tables

Based on $\pi$ = 0.4:

| | Predicted no AHD ($\hat{Y} = 0$) | Predicted AHD ($\hat{Y} = 1$) |
|---|---|---|
| Truly no AHD ($Y = 0$) | 93 | 71 |
| Truly AHD ($Y = 1$) | 38 | 101 |

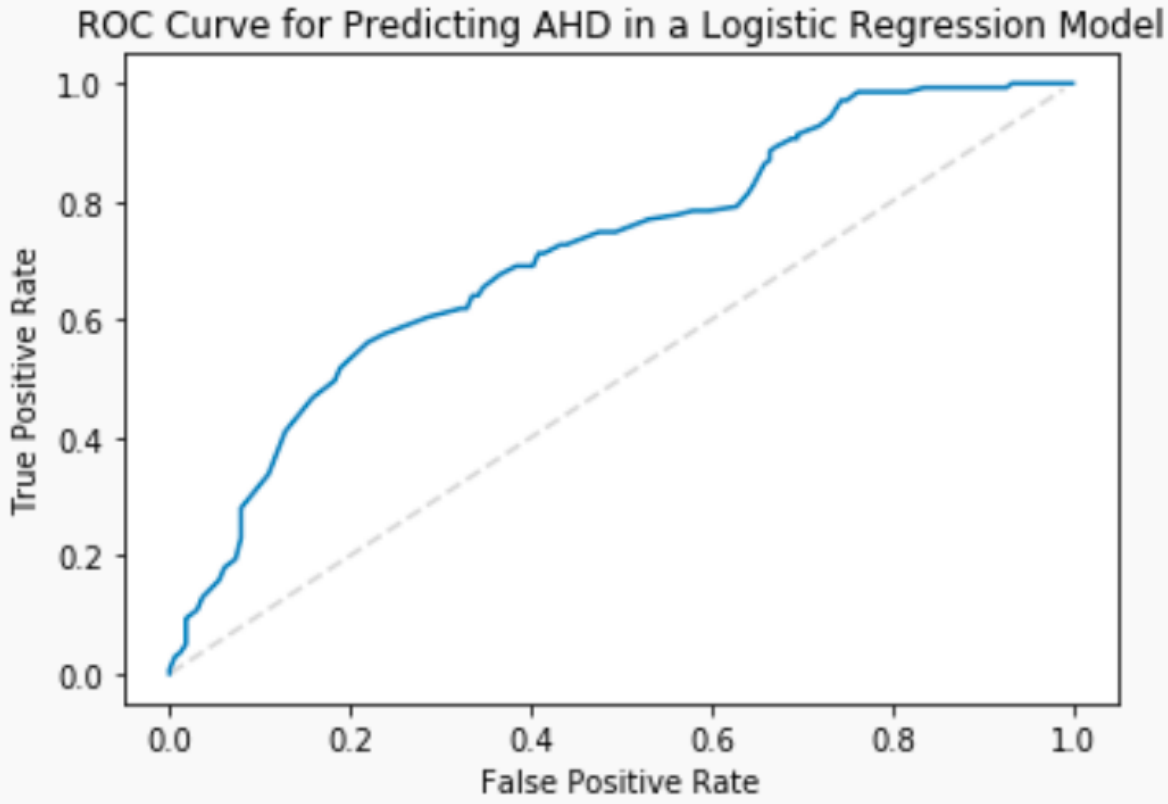
What has improved? What has worsened?

Based on $\pi$ = 0.6:

| | Predicted no AHD ($\hat{Y} = 0$) | Predicted AHD ($\hat{Y} = 1$) |
|---|---|---|
| Truly no AHD ($Y = 0$) | 138 | 26 |
| Truly AHD ($Y = 1$) | 74 | 65 |

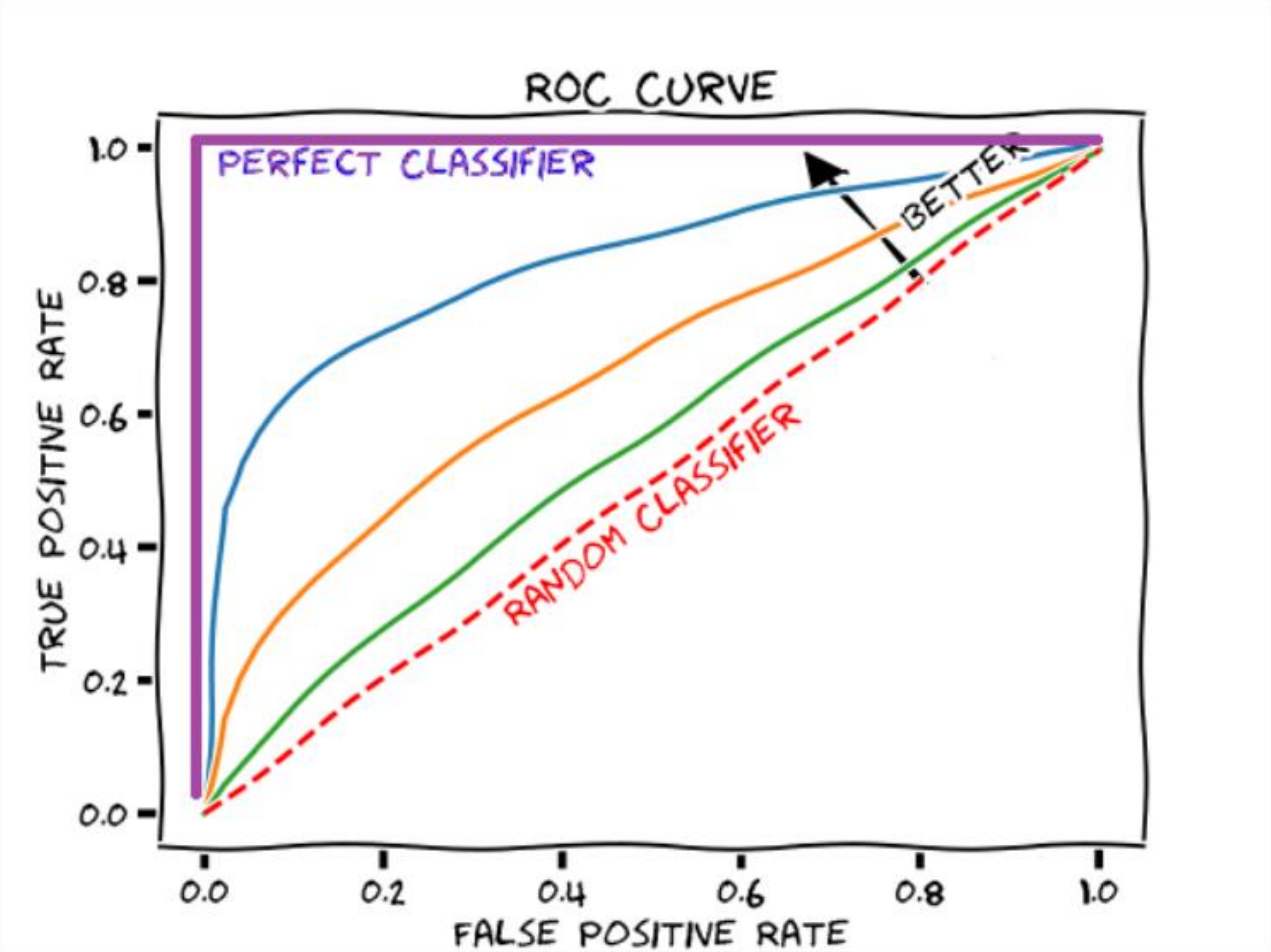
Which should we choose? Why?

# ROC Curves and Area-Under-the-Curve




What is the shape of an ideal ROC curve?

# ROC Curve Example



# AUC for measuring classifier performance

The overall performance of a classifier, calculated over all possible thresholds, is given by the **area under the ROC curve** (AUC).

An ideal ROC curve will hug the top left corner, so the larger the AUC the better the classifier.

This AUC then can be use to compare various approaches to classification: Logistic regression, *k*-NN, Decision Trees (to come), etc.